

Unified Theory Based on Time-Wave Dynamics (The P-model)

Part III-A: Quantum Foundations

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Abstract

This paper (Part III-A) explicitly separates what is fixed from first principles from what remains at the level of an effective theory in the quantum foundations of the P-model. While retaining the dynamical time-wave vector and the proper-time mapping fixed in Part I, it uniquely specifies the correspondence between the wave function and the time wave by requiring consistency with the Noether energy flow, and it organizes the minimum conditions under which the theory reduces to a Schrödinger-type form in the short-wavelength, nonrelativistic limit.

For the Born rule, the phase-diffusion relation

$$\Gamma_{\text{deph}} = \omega_*^2 \frac{k_B T_{\text{env}}}{\chi P} \tau_{\text{free}}$$

combined with the lowest-order linear detection response upgrades the detection probability $p \propto |\psi|^2$ to an **effective derivation under four conditions: phase mixing, linear detection response, weak backreaction, and frequentist coarse-graining**. For measurement, the detector-side stable state is taken as the reference, and the diagonal Lueders/Kraus update is recovered as a coarse-grained limit of decoherence and conditional update.

Electromagnetism is treated with a two-layer conclusion. The transverse component of the time-wave vector fixes the transverse-wave photon branch and the massless light mode, and it fixes the selection equation that chooses a single fine-structure constant from a single frozen action. At the same time, the origin of the local U(1) redundancy, charge quantization, and charge normalization remain outstanding tasks on the already adopted effective-theory side. Within this organization, the source dynamics of entanglement is fixed as three-wave mixing on the adopted U(1) sector. For quantum information, the paper reaches a minimal bridge that connects phase relaxation time, gate fidelity, pair-correlation retention depth, and the origin of error channels to existing results.

The point reached here is not to claim a replacement of quantum mechanics. Rather, it is to fix the physical contributions for which the P-model takes responsibility as reproducible equations, decision language, and audit procedures at the respective entry points of the Born rule, measurement, correlation, and quantum information.

1 Introduction

The stance the P-model can take toward quantum phenomena is to close the main interfaces of quantum mechanics with the equations of the P-model and thereby show how far the theory can be fixed as a falsifiable entry point. The P-model assumes time to be a wave, reinterprets particles as boundary conditions (reflections) of that wave, and can reinterpret correlations as the sharing of local P-structure. This paper, however, closes the major quantum-mechanical interfaces with P-model equations while leaving the empirical acceptance or rejection of each individual observational prediction to the verification judgments of Part III-B. For the weak interaction, the final observational judgment is fixed as "adopt Route A (transplant the electroweak effective theory) and reject Route B (the P-native replacement) by numerical audit," which restricts the scope to the operational verification of Route A. The theoretical mechanism candidates of Route B and the theoretical reference points for the W/Z masses from two-component coupled Q-balls are organized in Section 3.2 separately from the acceptance judgment itself. Quantum measurement is treated in the same way: this paper does not claim a complete derivation of the Born rule or state update. Instead, by combining phase diffusion and the linear detection response, it fixes the detection probability $p \propto |\psi|^2$ up to an **effective derivation under four conditions: phase mixing, linear detection response, weak backreaction, and frequentist coarse-graining**. State update is fixed as a coarse-grained effective derivation of the diagonal Lueders/Kraus update, while the remaining common residue is the probabilistic interpretation of single events and the irreversibility of statistical mechanics.

As the first entry point, Section 2.5.1 makes explicit that the minimum assumptions of the dynamical P sector from Part I reduce to a Schrödinger-type form in the short-wavelength, nonrelativistic limit, thereby fixing the minimum consistency conditions on the quantum side first.

What this paper handles is to fix, on the basis of public primary data, which assumption can fail under which analysis procedure, and to reduce that into differential predictions (falsification conditions). As in the astrophysical volume (Part II), the paper takes the verification summary (scoreboard) as the entry point and uses a format in which the inputs, processing conditions, and metrics of each verification can be tracked mechanically, so that future data additions (planned tests) can be absorbed on the same input/output surface.

This trilogy is intended to be read in the order **Part I -> Part II -> Part III**. Part I establishes the mapping, frozen quantities, and falsification form; Part II accumulates application-level tests from classical to cosmological scales within the same framework; and Part III-A / III-B treats the quantum domain on an isomorphic input/output surface, so that the unity of the P-model can be evaluated not as a "claim" but as the **consistency of its falsifiability**.

This organization of the quantum-foundations side also connects to the second and third attack routes. The exact vector hierarchy fixed on the mass-origin route yields a discrete series of particle-mass ratios, and the same late-universe background law of the P-model, $P_{\text{bg}}(t) \propto \exp[-H_0^{(P)}(t - t_0)]$, with $\omega_{\text{bg}} = H_0^{(P)}$, $\lambda_{\text{bg}} = 2\pi c/H_0^{(P)}$, $a_0 = cH_0^{(P)}/(2\pi)$, so the SPARC acceleration scale in Part II 4.14 is not read as a phenomenological fit constant but as the direct bridge quantity $\kappa_a = 1/(2\pi)$ connecting the mass-origin route and the background wave.

Style rule: the main text is closed only by the P-model's own definitional mapping procedure. Comparisons with reference frameworks (existing theories) are isolated in the reference notes at the end of sections, and the definitions or derivations in the main text are not replaced by those reference frameworks.

Reference note: this paper is not an attempt to reinterpret standard quantum mechanics (QM) by adding hidden variables. It starts from the minimum P-model equations and mappings fixed in Part I and tests how their wave-like properties reproduce quantum phenomena (interference and correlation)

as macroscopic observations. The standard formal structure of quantum mechanics is positioned as a phenomenological benchmark for checking the validity of the derived approximation formulas and metrics.

2 Theoretical Framework

The notation, minimum assumptions, and mappings used in this paper follow Part I. The re-statement of formulas is kept to a minimum; here only the minimum needed for standalone readability is summarized.

2.1 The P Field and the Potential ϕ

Let the static limit of the time-wave density be $P(x)$ and the far-field reference be P_∞ , adopting $P > P_\infty$ **in the vicinity of mass**. Systems with rotation and flow follow the rotational extension defined in Part I, $P_\mu = (P_t, P_1, P_2, P_3)$. Only the ratio P/P_∞ (dimensionless) appears in observables; the identification of the absolute value or dimension of P is outside the scope of this paper. However, P and P_∞ must have the same dimension in order for $\ln(P/P_\infty)$ to be defined. The gravitational potential is defined by

$$\phi \equiv -c^2 \ln \left(\frac{P}{P_\infty} \right)$$

which is zero at infinity and negative near mass.

2.2 Gravity (Equation of Motion)

The gravitational acceleration is unified as

$$\mathbf{a} = -\nabla\phi = c^2 \nabla \ln \left(\frac{P}{P_\infty} \right)$$

thereby fixing in equation form the statement that matter "slides down the P-gradient."

2.3 Clocks (Gravitational Term + Velocity Term)

The clock rate is decomposed as

$$\frac{d\tau}{dt} = \frac{P_\infty}{P} \left(\frac{d\tau}{dt} \right)_v$$

and the velocity factor is taken in the standard recovered form

$$\left(\frac{d\tau}{dt} \right)_v = \sqrt{1 - \frac{v^2}{c^2}}$$

Reference note: the velocity term in this section is isomorphic to the time-dilation factor of SR (special relativity) in the reference framework.

2.4 Light (Refraction Toward Higher P)

Light is assigned the refractive index

$$n(P) = \left(\frac{P}{P_\infty} \right)^{2\beta}$$

and, following Fermat's principle, is taken to **refract toward the higher-P side**. β is treated as the frozen value from Part I ($\beta \approx 1$); the main text of Parts II and III does not change the value of β case by case, and any update is concentrated into a revision of Part I.

Reference note: when the PPN framework is used as the reference frame, the coefficient for light can be compared under the correspondence $1 + \gamma = 2\beta$ (the factor of 2 is carried on the β side).

2.5 Quantum (Local P Structure, Correlation, and Selection)

Part I centers on gravity, clocks, and light propagation, but the starting point of the P-model, namely that "all matter is wave and its medium is time," can also connect to quantum phenomena such as interference and correlation. Here the paper does not claim a rederivation of quantum mechanics. Instead, it states on the theoretical side the **minimum stance and the verification entry point**: what is assumed and where the possible falsification points lie.

2.5.1 Fixing the Minimum Conditions (1-5) (P -> Phase -> Schr/KG)

To connect the Part I approach, which reduces dynamical P not to constants e, m but to $q(P), u(P)$, into the quantum domain, the following five minimum conditions are fixed. They function as necessary conditions for consistency with the Schrödinger equation.

1. **Wave description (complexification):** the rotation of a single P, $\theta \rightarrow \theta + \omega t$, is transferred into the complex representation ($e^{-i\omega t}$) as a phase on a 2D projective plane. This extends the interference pattern in spatial interference (C-I) from a scalar sum to a vector sum.
2. **Coupling between energy and frequency:** the angular rotation rate ω_P in the P-model is fixed to be proportional to the observed energy E through $E = \hbar\omega_P$. This guarantees consistency with the local rate of time defined by the clock mapping.
3. **Coupling between momentum and wavelength:** from the relation between the interference path Δx and phase, the wave number k of a traveling wave is fixed to be proportional to momentum through $p = \hbar k$.
4. **Energy dispersion relation (nonrelativistic limit):** to define the local traveling state of P, the classical energy conservation law $E = \frac{p^2}{2m} + V$ in the $v \ll c$ regime is accepted. In the P-model, V is described as spatial distortion (refraction) due to an external field.
5. **Principle of superposition (linearity):** it is assumed that, in the collective behavior of P, a linear combination of multiple states is itself a state.

From these five fixed conditions, differentiating $\psi(x, t) = Ae^{i(kx - \omega t)}$ once in time and twice in space yields the form

$$i\hbar \frac{\partial \psi}{\partial t} = \left(-\frac{\hbar^2}{2m} \nabla^2 + m\phi \right) \psi \quad (1)$$

The minimum relativistic free-field form is the Klein-Gordon equation

$$\left(-\frac{1}{c^2} \frac{\partial^2}{\partial t^2} + \nabla^2 - \frac{m^2 c^2}{\hbar^2} \right) \Psi = 0 \quad (2)$$

This is isomorphic to the usual derivation process of the Schrödinger equation, but in the P-model the difference is that ψ is treated as a "geometric mapping of the local rotation and propagation of P."

Here, the minimal form of the $\psi \leftrightarrow P$ correspondence is fixed under the following conditions.

First, on top of the static background solution $P_*(x) > 0$ from Part I, we superpose a small fluctuation on the positive-frequency branch,

$$P(x, t) = e^{-i\omega_* t} (P_*(x) + \delta P_+(x, t)) + e^{+i\omega_* t} \delta P_-(x, t), \quad \left| \frac{\delta P_{\pm}}{P_*} \right| \ll 1$$

where the positive-frequency branch δP_+ is taken to correspond to particle-like excitations, while δP_- is relegated to higher-order corrections in the nonrelativistic, weak-field, slowly varying branch. Then

$$\delta P_+(x, t) = P_*(x) a(x, t) e^{i\vartheta(x, t)}$$

if we write it this way, a is the dimensionless amplitude fluctuation relative to the background, and ϑ is the local phase with the carrier removed. Accordingly,

$$\psi(x, t) \equiv \sqrt{2\omega_* P_*^2} \frac{\delta P_+(x, t)}{P_*(x)} = \sqrt{2\omega_* P_*^2} a(x, t) e^{i\vartheta(x, t)}$$

is adopted as the **minimal definition of the wave function in the P-model**. With this definition,

$$|\psi| \propto \left| \frac{\delta P_+}{P_*} \right|, \quad \arg \psi = \vartheta$$

so the amplitude of ψ corresponds to the relative fluctuation of the positive-frequency envelope with respect to the background field P_* , and its phase corresponds to the residual phase obtained by subtracting the carrier from the local rotation of the time wave.

That this normalization is not arbitrary can be checked by reducing the Noether conservation law for time translation fixed in Part I 2.7.0,

$$\partial_\mu T_{\text{tot}}^{\mu\nu} = 0$$

to second-order perturbations on a static background. Isolating only the second-order energy flow gives

$$\partial_\mu \delta T_{(2)}^{\mu 0} = 0$$

and under the positive-frequency, slowly varying approximation $|\partial_t \psi| \ll \omega_* |\psi|$,

$$\delta T_{(2)}^{00} = \hbar \omega_* |\psi|^2 + \frac{\hbar^2}{2m_*} |\nabla \psi|^2 + m_* \phi |\psi|^2 + \mathcal{O}\left(\varepsilon_{\text{ad}}^2, \left| \frac{\delta P_+}{P_*} \right|^3\right), \quad m_* c^2 = \hbar \omega_*$$

$$\delta T_{(2)}^{i0} = \frac{\hbar^2}{2m_* i} (\psi^* \partial_i \psi - \partial_i \psi^* \psi) + \mathcal{O}\left(\varepsilon_{\text{ad}}^2, \left| \frac{\delta P_+}{P_*} \right|^3\right)$$

we obtain

$$\partial_t (\hbar \omega_* |\psi|^2) + \nabla \cdot \mathbf{S}_\psi = 0, \quad \mathbf{S}_\psi \equiv (\delta T_{(2)}^{10}, \delta T_{(2)}^{20}, \delta T_{(2)}^{30})$$

Therefore, $|\psi|^2$ is fixed uniquely, at least within this approximation, not as a probability density but as the normalization of the wave-packet energy density transported on the background. In other words, the gate for the $\psi \leftrightarrow P$ correspondence closes by reading the relative fluctuation of the positive-frequency branch with a normalization that is consistent with the Noether energy flow.

Returning this definition to the positive-frequency branch of the KG equation, the envelope equation on a static background becomes

$$i\hbar \frac{\partial \psi}{\partial t} = \left(-\frac{\hbar^2}{2m_*} \nabla^2 + m_* \phi + U_{\text{lin}}(P_*) \right) \psi$$

Here $U_{\text{lin}}(P_*)$ can be absorbed into a redefinition of the carrier set by the background solution, so in Part III the Schrödinger form

$$i\hbar \frac{\partial \psi}{\partial t} = \left(-\frac{\hbar^2}{2m_*} \nabla^2 + m_* \phi \right) \psi$$

is taken as the canonical form.

Points fixed in this paper:

1. The amplitude of ψ is the normalized form of $\delta P_+/P_*$ and reads the relative envelope fluctuation

from the background.

2. The phase of ψ corresponds to the local phase ϑ with the carrier removed.
3. The normalization of ψ is fixed to remain consistent with the Noether energy flow $\partial_\mu \delta T_{(2)}^{\mu 0} = 0$, rather than being left as an arbitrary constant.
4. The effective range is limited to a static background, linear perturbations, the positive-frequency branch, and the weak-field, slowly varying regime.

Still not derived here (and therefore not claimed): Not derived here 1: a **complete** first-principles derivation of the Born rule (probability density $|\psi|^2$), including phase randomization, linear detection response, and conditional update closed from P dynamics all the way through detector irreversibility. Not derived here 2: a first-principles derivation of measurement (state update), including a microscopic description of device irreversibility and amplification. Not derived here 3: spin, charge, gauge fields, and interaction terms (electromagnetism onward is introduced minimally in later sections).

Observability of P-model-specific corrections (entry point): Observability 1: the "form" of the Schrödinger equation is isomorphic to the standard one, but because the definition of the potential ϕ is specific to the P-model, differential predictions can appear when P deviates from the standard gravitational potential in the strong-field regime (while remaining equivalent in the weak field). Observability 2: experimentally, the main entry point is atom interferometry that involves not only matter phases but also **light (laser phase and readout)**, where phase terms from light propagation (through β) can generate differential predictions.

2.5.2 Born Rule: Domain of Adoption and Regimes of Deviation (P-model Stance)

This paper does not claim a complete first-principles derivation of the **entire** Born rule. In particular, the probabilistic interpretation of single events and irreversibility in the thermodynamic limit remain as common residues, not P-model-specific ones. At the same time, the A1/A2 organization in this paper upgrades the detection probability $p \propto |\psi|^2$ itself to an **effective derivation under four conditions: phase mixing, linear detection response, weak backreaction, and frequentist coarse-graining**. On that basis, this section makes explicit both the domain in which the rule is adopted and the conditions under which the Born rule, as an approximation in use, can appear to fail.

Adopted here (fixed): Adoption A (position measurement): the detection probability density is treated as

$$p(x) \propto |\psi(x)|^2 \quad (3)$$

Adoption B (continuous detection): the local detection rate is approximated as

$$\lambda(x, t) = \kappa(x, t) |\psi(x, t)|^2$$

where κ is a coarse-grained coefficient for the detector response, threshold, and gain.

Statistical entry point (initial bridge): Under the $\psi \leftrightarrow P$ correspondence fixed above, ψ is the normalized form of the relative fluctuation $\delta P_+/P_*$ of the positive-frequency envelope, and $|\psi|^2$ can be read as a proxy for local transported energy density consistent with the Noether energy flow. Let the pre-detection amplitude entering coarse-grained cell j be

$$\mathcal{A}_j = \sum_{n=1}^{N_j} a_{jn} e^{i\theta_{jn}}, \quad \sum_{n=1}^{N_j} |a_{jn}|^2 \equiv \rho_j \propto |\psi_j|^2$$

Here θ_{jn} is the microscopic phase with the carrier removed, and if the fluctuations of many time-wave modes, environmental modes, and trial-to-trial variations satisfy

$$\langle e^{i(\theta_{jn}-\theta_{jm})} \rangle = \delta_{nm}$$

then

$$\langle |\mathcal{A}_j|^2 \rangle = \sum_{n,m} a_{jn} a_{jm}^* \langle e^{i(\theta_{jn}-\theta_{jm})} \rangle = \sum_n |a_{jn}|^2 = \rho_j$$

and the cross terms are washed out in the ensemble average. If, furthermore, the single-event detection response is linear in the local transported energy,

$$\lambda_j = \kappa_j \rho_j$$

and if κ_j is spatially constant or can be removed as a known factor by flat-field calibration, then

$$f_j = \frac{\lambda_j}{\sum_k \lambda_k} = \frac{\rho_j}{\sum_k \rho_k} \propto |\psi_j|^2$$

Therefore $|\psi|^2$ appears as the **frequentist limit of the normalized envelope energy density**.

A1 (origin of phase diffusion): Let the time component of the static background be

$$P_t(x, t) = P_*(x) [1 + u_{\text{env}}(x, t)], \quad |u_{\text{env}}| \ll 1$$

Using the proper-time mapping fixed in Part I 2.7.0,

$$\frac{d\tau}{dt} = \frac{P_{\text{ref}}}{P_t} \sqrt{1 - v^2/c^2}$$

and expanding to first order in u_{env} after removing the mean carrier term $\omega_* \tau_*$, we obtain

$$\dot{\vartheta}(t) = -\omega_* u_{\text{env}}(t)$$

Thus the phase-mixing problem reduces to an open-system problem in which the local phase ϑ after carrier removal is driven by the low-frequency background fluctuation u_{env} .

In the Mori-Zwanzig coarse-graining of Part I 2.7, the decorrelation time of the same background branch is fixed as $\tau_{\text{free}} = \Gamma_{\text{path}}^{-1}$. Taking the minimal thermal closure as

$$\dot{u}_{\text{env}} = -\Gamma_{\text{path}} u_{\text{env}} + \xi_{\text{env}}(t), \quad \langle \xi_{\text{env}}(t) \xi_{\text{env}}(0) \rangle = 2\Gamma_{\text{path}} \frac{k_B T_{\text{env}}}{\chi_P} \delta(t)$$

we identify χ_P with the same coarse-grained susceptibility defined in Part I 2.7 through

$$\Gamma_{\text{path}} = \frac{g_P^2}{\chi_P} \int_0^\infty \langle \delta J_x(t) \delta J_x(0) \rangle dt$$

and since T_{env} is an external condition on the background, it is not a new free parameter. Then

$$\langle u_{\text{env}}(t)u_{\text{env}}(0) \rangle = \frac{k_B T_{\text{env}}}{\chi_P} e^{-\Gamma_{\text{path}}|t|}$$

follows.

$$\text{Since } \Delta\vartheta(T) = -\omega_* \int_0^T u_{\text{env}}(s) ds,$$

$$\text{Var}[\Delta\vartheta(T)] = 2\omega_*^2 \frac{k_B T_{\text{env}}}{\chi_P} \left[\frac{T}{\Gamma_{\text{path}}} - \frac{1 - e^{-\Gamma_{\text{path}}T}}{\Gamma_{\text{path}}^2} \right]$$

Under the Gaussian/Markov approximation, the off-diagonal coherence is given by

$$\langle e^{i\Delta\vartheta(T)} \rangle = \exp \left[-\frac{1}{2} \text{Var}[\Delta\vartheta(T)] \right]$$

and for $\Gamma_{\text{path}}T \gg 1$,

$$\langle e^{i\Delta\vartheta(T)} \rangle \simeq e^{-\Gamma_{\text{deph}}T}, \quad \Gamma_{\text{deph}} = \omega_*^2 \frac{k_B T_{\text{env}}}{\chi_P} \tau_{\text{free}}$$

Furthermore, using the transport relation from Part I 2.7,

$$\tau_{\text{free}} = \frac{L_{\text{path}}}{c_w} = \left(\frac{v}{L_{\text{corr}}} + A_{\text{col}} \rho T_{\text{env}}^{-3/2} \right)^{-1}$$

we obtain

$$\Gamma_{\text{deph}} = \omega_*^2 \frac{k_B T_{\text{env}}}{\chi_P} \frac{1}{\frac{v}{L_{\text{corr}}} + A_{\text{col}} \rho T_{\text{env}}^{-3/2}} = \omega_*^2 \frac{k_B T_{\text{env}}}{\chi_P} \frac{1}{\Gamma_{\text{path}}}$$

which closes the expression. Therefore the diagnostic quantity for phase mixing is

$$\Gamma_{\text{deph}} T_{\text{obs}} = (\omega_* \tau_{\text{free}})^2 \left(\frac{k_B T_{\text{env}}}{\chi_P} \right) \left(\frac{T_{\text{obs}}}{\tau_{\text{free}}} \right)$$

If this is $\gg 1$, then the condition required for the statistical bridge,

$$\langle e^{i(\theta_{jn} - \theta_{jm})} \rangle \rightarrow \delta_{nm}$$

holds after coarse-graining. Conversely, if it is $\lesssim 1$, cross terms can survive in ultracold, highly isolated systems or over short observation windows. This connects directly to the "regimes of possible deviation" described below.

A2 (origin of the linear detection response): Take the minimal interaction fixed in Part I 2.7.0,

$$\mathcal{L}_{\text{int}} := g_P P_\mu J_{\text{matter}}^\mu$$

and expand it around the detector's macroscopic background

$$P_{\mu, \text{det}}(x, t) := P_{\mu, \text{det}, *}(x) + \delta P_{\mu, \text{det}}(x, t)$$

and the incoming positive-frequency signal

$$\delta P_+(x, t) := P_*(x_j) \psi_j(t) u_j(x) e^{-i\omega_* t}, \quad \int_{V_j} |u_j(x)|^2 d^3x = 1$$

Here V_j is the position-readout cell, u_j is the mode function within the cell, and $|\delta P_+/P_{\text{det},*}| \ll 1$ is taken as the weak-signal gate. Since the stationary background $P_{\mu, \text{det},*}$ itself does not induce transitions, the lowest-order term that drives detection is, with respect to the detector-current fluctuation δJ_{det}^0 ,

$$H_{\text{int},j}^{(1)}(t) := -g_P \int_{V_j} d^3x [\delta P_+(x, t) \delta J_{\text{det}}^0(x, t) + \delta P_+^*(x, t) \delta J_{\text{det}}^0(x, t)]$$

The detector's initial pointer-ready state is denoted by $|D_0\rangle$, and the microscopic final state after absorption by $|D_\alpha\rangle$. Then Fermi's golden rule gives

$$\lambda_{j \rightarrow \alpha} := \frac{2\pi g_P^2 P_*(x_j)^2}{\hbar} |\psi_j|^2 \left| \left\langle D_\alpha \left| \int_{V_j} d^3x u_j(x) \delta J_{\text{det}}^0(x) \right| D_0 \right\rangle \right|^2 \delta(E_\alpha - E_0 - \hbar\omega_*)$$

The key point is that the lowest-order transition amplitude is **linear** in $\delta P_+ \propto \psi_j$, so in the transition rate $|\psi_j|^2$ factors out in front. Summing over final states then gives

$$\lambda_j := \sum_{\alpha} \lambda_{j \rightarrow \alpha} := \kappa_j |\psi_j|^2$$

with

$$\kappa_j := \frac{2\pi g_P^2 P_*(x_j)^2}{\hbar} \sum_{\alpha} \left| \left\langle D_\alpha \left| \int_{V_j} d^3x u_j(x) \delta J_{\text{det}}^0(x) \right| D_0 \right\rangle \right|^2 \delta(E_\alpha - E_0 - \hbar\omega_*)$$

giving the detector-side local spectral response. If the weight of the coarse-grained cell is read as

$$\rho_j := \int_{V_j} |\psi(x)|^2 d^3x$$

the same expression can be written as

$$\lambda_j := \bar{\kappa}_j \rho_j$$

where $\bar{\kappa}_j$ is the detector spectral density averaged over the cell and can be treated as known after flat-field calibration. As an equivalent correlation-function representation, define the operator

$$\hat{\mathcal{O}}_j := \int_{V_j} d^3x u_j(x) \delta J_{\text{det}}^0(x)$$

Then

$$S_{\mathcal{O}\mathcal{O}}^{(j)}(\omega) := \int_{-\infty}^{\infty} dt e^{i\omega t} \langle D_0 | \hat{\mathcal{O}}_j(t) \hat{\mathcal{O}}_j(0) | D_0 \rangle$$

implies

$$\kappa_j := \frac{g_P^2 P_*(x_j)^2}{\hbar^2} S_{\mathcal{O}\mathcal{O}}^{(j)}(\omega_*)$$

Thus the only additional ingredients required in A2 are the coupling g_P already fixed in Part I and the detector-current correlation; no new free parameter of the P-model is introduced. Next, if one keeps the next nonlinear term around the detector background, the amplitude acquires one extra factor of $\delta P_+/P_{\text{det},*}$. Therefore the relative correction to the rate is suppressed as

$$\frac{\Delta\lambda_j^{(\text{nl})}}{\lambda_j^{(1)}} = \mathcal{O}\left(\left|\frac{\delta P_+}{P_{\text{det},*}}\right|^2\right)$$

In the asymmetric limit where the detector is a macroscopic background and the incoming wave is a small fluctuation, $|\delta P_+/P_{\text{det},*}|^2 \ll 1$, so the linear response is stable as the lowest-order consequence. After removing κ_j by flat-fielding,

$$f_j^{(\text{ff})} := \frac{\lambda_j/\kappa_j}{\sum_k \lambda_k/\kappa_k} = \frac{\rho_j}{\sum_k \rho_k} \propto |\psi_j|^2$$

Thus the A2 gap closes in the form: "in the weak-signal / weak-backreaction limit, the detector's lowest-order transition rate is proportional to $|\psi|^2$."

What can and cannot be claimed at this stage: From the derivation above, among the previously open items of the Born rule:

1. Why phase mixing generally occurs

is closed as an open-system closure using τ_{free} , Γ_{path} , and χ_P from Part I 2.7.

1. Why the detector response may be treated as linear

is closed by the minimal coupling $\mathcal{L}_{\text{int}} = g_P P_\mu J^\mu$ from Part I 2.7.0 together with the weak-signal gate $|\delta P_+/P_{\text{det},*}|^2 \ll 1$. Therefore the detection probability $p \propto |\psi|^2$ and local detection rate $\lambda_j = \bar{\kappa}_j \rho_j$ reach an **effective derivation under phase mixing, lowest-order linear response, weak backreaction, and frequentist coarse-graining**. Meanwhile,

1. Why conditioning (state update) takes the Lueders / Kraus form

is closed up to an **effective derivation** through decoherence plus conditioning in Sections 2.5.3 and 2.5.4. Accordingly, the present judgment is: **"the Born-rule detection probability is fixed as an effective derivation under phase mixing, linear detection response, weak backreaction, and frequentist coarse-graining, while diagonal Lueders/Kraus update is fixed as a coarse-grained effective derivation."** The remaining common residues are: (i) foundational probability theory concerning how probabilities are assigned to single events, and (ii) C3, namely why this coarse-grained update appears thermodynamically irreversible.

Domain of adoption (conditions under which validity is expected): Condition 1: the phase-diffusion gate $\Gamma_{\text{deph}} T_{\text{obs}} \gg 1$ holds, so cross terms are washed out in the ensemble average. Supplement (justification of the Markov approximation): the OU closure in A1 is applied to the local phase ϑ after removal of the carrier. Therefore, the correlation time

$$\tau_{\text{corr}} = \Gamma_{\text{path}}^{-1} = \tau_{\text{free}}$$

should not be compared with the raw carrier period

$$\tau_{\text{carrier}} = \omega_*^{-1}$$

but with

$$T_{\text{obs}}$$

which defines the coarse-graining of counting, readout, and the envelope. The ratio required by the Markov closure is in fact

$$\frac{\tau_{\text{corr}}}{T_{\text{obs}}} = \frac{1}{\Gamma_{\text{path}} T_{\text{obs}}} = \frac{\tau_{\text{free}}}{T_{\text{obs}}} \ll 1$$

which reduces to the same coarse-grained criterion as the phase-diffusion gate in A1. By contrast,

$$\frac{\tau_{\text{corr}}}{\tau_{\text{carrier}}} = \omega_* \tau_{\text{free}}$$

is only a diagnostic ratio comparing directly to the optical carrier, and it is not a condition that invalidates the open-system closure of this section where the carrier has already been factored out. Taking the representative optical readout condition $\omega_* = 3.0 \times 10^{15} \text{ s}^{-1}$, together with $\tau_{\text{free}} = 10^{-13} \dots 10^{-10} \text{ s}$ allowed by the transport relation in Part I 2.7 and the readout windows $T_{\text{obs}} = 10^{-9} \dots 10^{-5} \text{ s}$, the coarse-graining ratios are

Representative condition	$\omega_* \text{ (s}^{-1}\text{)}$	$\tau_{\text{free}} \text{ (s)}$	$T_{\text{obs}} \text{ (s)}$	$\omega_* \tau_{\text{free}}$	$\tau_{\text{free}}/T_{\text{obs}}$
Optical high-speed bin readout	3.0×10^{15}	1.0×10^{-13}	1.0×10^{-9}	3.0×10^2	1.0×10^{-4}
Optical ns readout	3.0×10^{15}	1.0×10^{-12}	1.0×10^{-7}	3.0×10^3	1.0×10^{-5}
Optical us integration	3.0×10^{15}	1.0×10^{-10}	1.0×10^{-5}	3.0×10^5	1.0×10^{-5}

and in all cases $\Gamma_{\text{path}} T_{\text{obs}} = 10^4 \dots 10^5 \gg 1$. Therefore, for representative room-temperature optical readout windows, the Markov approximation is well justified on the coarse-grained detector timescale. Representative reproducibility artifacts are listed in the Part IV ledger. Condition 2: the device remains in the weak-signal linear-response regime ($|\delta P_+ / P_{\text{det},*}|^2 \ll 1$, and κ does not depend strongly on the intensity, phase, or past history of ψ). Condition 3: the Markov approximation holds (the device/environment correlation time is sufficiently shorter than the system timescale). Condition 4: backreaction is negligible (during one trial, the energy flow associated with detection does not significantly change the $u(P)$ field itself). Supplement (accumulated backreaction under repeated detections): the A2 weak-signal gate for a single event requires

$$\left| \frac{\delta P_+}{P_{\text{det},*}} \right|^2 \ll 1$$

but if N detections accumulate on the same detector background, the minimal estimate becomes

$$\frac{\Delta P_{\text{det},*}}{P_{\text{det},*}} \sim N \left| \frac{\delta P_+}{P_{\text{det},*}} \right|^2$$

Therefore the perturbative gate including many events is

$$N \left| \frac{\delta P_+}{P_{\text{det},*}} \right|^2 \ll 1$$

Representative examples are summarized in the Part IV reproducibility ledger; only the results are recorded here.

Representative condition	$ \delta P_+/P_{\text{det},*} $	N	$N \delta P_+/P_{\text{det},*} ^2$
single-photon counting	10^{-20}	10^{10}	10^{-30}
bright calibrated run	10^{-10}	10^{12}	10^{-8}
ultra-intense many-body injection	10^{-4}	10^8	1

Thus, while the cumulative change of the detector background can be neglected for single-photon detection or calibrated many-shot runs, regimes such as ultra-intense laser irradiation or many-body injection into a Bose-Einstein condensate can approach

$$N \left| \frac{\delta P_+}{P_{\text{det},*}} \right|^2 \gtrsim 1$$

In such a regime, the linear response of A2 should not be used unchanged; one must switch to a different specification that includes background retuning and nonlinear detector dynamics. Condition 5: the system is in the weak field ($|\phi|/c^2 \ll 1$), so higher-order terms of the clock mapping (the exponential nonlinearity) remain sufficiently small relative to the experimental precision.

Regimes where deviations may occur (domains requiring an additional specification):

Regime 1 (macroscopic superposition / highly isolated systems / many-body injection): if the system becomes large, or if $\Gamma_{\text{deph}} T_{\text{obs}} \lesssim 1$ in an ultracold, high- Q , short-observation-window setting, then one or more of (i) phase mixing, (ii) neglect of backreaction, (iii) the Markov approximation, and (iv) linear response may fail. A concrete example is when $N |\delta P_+/P_{\text{det},*}|^2 \gtrsim 1$ under ultra-intense laser irradiation or many-body injection into a Bose-Einstein condensate, in which case the adopted form of the Born rule itself may need to be changed. Regime 2 (strong gravity): in regions where $|\phi|/c^2$ is not negligible, the nonlinearity, spacetime dependence, and fluctuations of u become explicit, and the detection-rate coefficient κ may itself depend on u (currently left undefined). Regime 3 (self-gravitating mean field): if one **feeds the quantum mass density into the source of u** (for example $\rho \rightarrow m|\psi|^2$), then the Schrödinger equation becomes of the Schrödinger-Newton type (nonlinear), and the visibility, phase, and stability of interference/superposition change. This is an extra assumption beyond the Part I core, and if adopted it should be fixed explicitly as a "separate hypothesis" with its own falsification conditions. Space-based macroscopic interferometer plans such as MAQRO-like concepts could become the decisive tests.

Observability (quantitative form): Observable A: if a small deviation of the adopted form of the Born rule is written as $p \rightarrow p(1 + \delta)$, then in simple counting statistics one obtains

$$|\delta| \gtrsim \frac{3}{\sqrt{N}}$$

which gives a 3σ detection threshold, where N is the effective number of trials. Observable B: if κ

responds linearly to u (or ∇u), so that $\delta \approx \epsilon_u |u|$ or $\delta \approx \epsilon_g \ell |\nabla u|$, then

$$u = -\phi/c^2, \quad |\nabla u| = |\nabla \phi|/c^2$$

so in the laboratory gravitational field ($|\nabla \phi| \approx g$), one has $|u| \sim 10^{-9}$ and $|\nabla u| \sim 10^{-16} \text{ m}^{-1}$, both extremely small. Therefore any effect large enough to break the Born rule would require **another amplification mechanism, such as device nonlinearity, self-gravity, or non-Markovianity**. Observable C: the scale at which self-gravity (mean field) becomes relevant in macroscopic interference is estimated, in terms of wave-packet size R and interference time T , by

$$\frac{E_G}{\hbar} T \sim \frac{Gm^2}{\hbar R} T \gtrsim 1 \quad (4)$$

where $E_G \sim Gm^2/R$ is a rough scale for the gravitational self-energy. The mass, size, and time at which an "additional effect becomes visible" are set by this kind of dimensionless quantity. For example,

$$m_{\text{crit}} \sim \sqrt{\frac{\hbar R}{GT}}$$

is the scale at which the gravitational self-effect contributes of order 1 radian to the phase. For $R = 100 \text{ nm}$ and $T = 1 \text{ s}$, this gives $m_{\text{crit}} \sim 4 \times 10^{-16} \text{ kg}$ (about $2 \times 10^{11} \text{ amu}$).

Experimental connection (predictions for planned experiments): Concrete examples of m_{crit} ($m_{\text{crit}} \equiv \sqrt{\hbar R/(GT)}$, rough estimate) are:

R (wave-packet size)	T (interference time)	m_{crit}	Corresponding mass scale
1 nm	1 ms	$1.3 \times 10^{-15} \text{ kg}$ ($7.6 \times 10^{11} \text{ amu}$)	Upper scale of sub-micron to micron nanoparticle interference
100 nm	1 s	$4.0 \times 10^{-16} \text{ kg}$ ($2.4 \times 10^{11} \text{ amu}$)	Micron-class nanoparticles (10^{11} amu scale)
1 μm	10 s	$4.0 \times 10^{-16} \text{ kg}$ ($2.4 \times 10^{11} \text{ amu}$)	Same R/T ratio
10 μm	100 s	$4.0 \times 10^{-16} \text{ kg}$ ($2.4 \times 10^{11} \text{ amu}$)	Same R/T ratio

Space interferometer concepts such as MAQRO can simultaneously realize long T and large R , so they are relatively more likely to approach m_{crit} scales of the kind listed above while suppressing environmental decoherence better than ground-based systems. If one adopts the additional hypothesis that quantum mass density enters the source of u (a Schrödinger-Newton-type mean field), then phase shifts and visibility loss may be predicted near $m \sim m_{\text{crit}}$, appearing as apparent deviations from the Born rule.

By contrast, in molecular interference such as OTIMA ($10^6 \dots 10^8 \text{ amu}$ scale), one has $m \ll m_{\text{crit}}$, so this paper does not predict a Born-rule deviation there. Altitude-comparison optical lattice clocks also predict $\epsilon \approx 0$ in the weak field and remain consistent with current observations (Section 4.3).

A direct test of Born-rule deviation is therefore difficult with present ground-based experiments, and future space experiments, including MAQRO-like concepts, could become decisive.

Supplement (boundary of the single-event probability interpretation): what is fixed directly by the

derivation in this section is the coarse-grained law for the frequentist limit of independent trials,

$$f_j \rightarrow p_j \propto |\psi_j|^2 \quad (N \rightarrow \infty)$$

The interpretation of a single event, namely "why this result occurs with this probability in one measurement," belongs to the foundations of probability theory under any of the frequentist, Bayesian, or de Finetti viewpoints, and it remains a common residue across Copenhagen, many-worlds, QBism, and other interpretations of quantum mechanics. Therefore what remains here is not a P-model-specific underivation, but a **common residue concerning the foundations of single-event probability**. With that boundary made explicit, the Born rule is reclassified here not as a "conditional derivation" but as an **effective derivation under the four conditions of phase mixing, linear detection response, weak backreaction, and frequentist coarse-graining**.

Reference note (relation to Gleason-type structure): The physical route of the P-model (A1 phase diffusion + A2 linear detection) and Gleason's structural route (Hilbert-space linearity + noncontextuality) are independent. Once the fifth condition in Section 2.5.1 assumes the linear structure of superposition, part of Gleason's premises is already satisfied. However, this paper does not push uniquely from there to a probability measure; instead it derives

$$p \propto |\psi|^2$$

from the lowest-order transition rate of the detector coupling. Therefore the fact that both routes reach the same

$$|\psi|^2$$

already serves as a double guarantee of the robustness of the derivation. In particular, the answer to "why not

$$|\psi|^4$$

?" is that in A2 the transition amplitude is first order in ψ and the rate is given by its absolute square, while higher-order processes are suppressed by

$$\left| \frac{\delta P_+}{P_{\text{det},*}} \right|^2 \ll 1, \quad N \left| \frac{\delta P_+}{P_{\text{det},*}} \right|^2 \ll 1$$

the gates above.

2.5.3 Positioning of Measurement (State Update): Lueders Adoption, Decoherence, and Penrose-Diosi

This paper does not assert "wave-function collapse" as a literal physical fact. Instead it treats the measurement result m as a macroscopic record and state update as conditioning on that record, namely an effective state conditioned by the observed record. In that minimal form, it **adopts** both projective measurement (Lueders type) and general measurement (POVM/Kraus), while stating clearly the reasons for adoption and the points that remain unresolved.

Adopted here (fixed): Adoption A (projective measurement; Lueders):

$$p(m) = \text{Tr}(\Pi_m \rho), \quad \rho \rightarrow \rho_m = \frac{\Pi_m \rho \Pi_m}{\text{Tr}(\Pi_m \rho)}$$

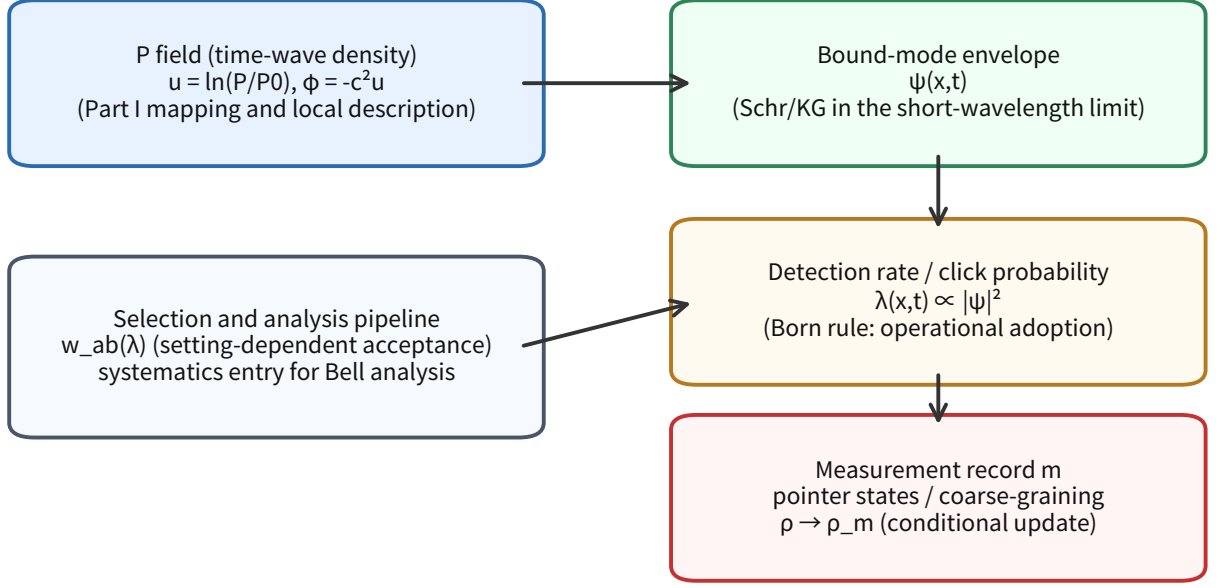
Adoption B (general measurement; Kraus):

$$p(m) = \text{Tr}(E_m \rho), \quad \rho \rightarrow \rho_m = \frac{M_m \rho M_m^\dagger}{\text{Tr}(E_m \rho)}, \quad E_m = M_m^\dagger M_m, \quad \sum_m E_m = I$$

Reasons for adoption (why Lueders is used): Reason 1: it is the minimal update rule and is consistent with (i) conditioning by records, (ii) reproducibility under immediate remeasurement (the same record is stably repeated), and (iii) the standard open-system formalism (quantum trajectories and Lindblad unraveling). Reason 2: in the vocabulary of the P-model, Π_m and M_m are not "arbitrary axioms" but are interpreted as **equivalence classes of macroscopic records (pointer states)** that remain after coarse-graining over the many degrees of freedom of the apparatus and environment. Accordingly, the update is treated as conditional update, not as a literal assertion of collapse.

Decoherence mechanism (phrased in P-model language): Mechanism 1: measurement is the interference between the local P-structure (mode) of the target and the large P-structure of the apparatus (boundary condition + dissipation + amplification), and in the effective description after coarse-graining, the off-diagonal components in the pointer basis decay (decoherence). Mechanism 2: in the C1/C2 organization of this paper, the pointer basis is fixed as the stable mode of the P background, and the Π_m / Kraus operators are fixed as conditionings on the detector-record region F_r . What remains open at this stage is the statistical-mechanical question of why the environmental trace appears irreversible; the update rule itself has been closed up to an effective derivation. Figure~1 fixes that placement, while detailed discussion is separated into supplementary material.

Quantum Measurement Positioning (Born Rule and State Update)



This first release fixes the boundary between derived elements and adopted elements, and makes the verification entry explicit against the criticism that the model is only semi-classical. A first-principles derivation of the Born rule and update rule remains future work.

Figure 1: This figure summarizes the placement of the Born rule ($|\psi|^2$) and state update (conditional update).

Relation to Penrose-Diosi-type proposals: Relation 1: this paper does not adopt objective collapse, namely the hypothesis that gravity spontaneously collapses the wave packet. Therefore the Penrose-Diosi type remains a **separate hypothesis** and is not identified with the present framework. Relation 2: if one later introduces objective collapse starting from fluctuations of P or backreaction, it must be constrained by primary data on the same footing as existing Penrose-Diosi / CSL bounds, and its domain of adoption and falsification conditions must be fixed separately. That is left undefined in this paper.

Falsification entry point: selection and the statistical population: Entry definition: in Bell tests, observational procedures (timestamp coincidence windows, trial-based definitions, and event-ready windows) can change the **set of events admitted into the statistical population**. Writing this as a selection weight $w_{ab}(\lambda)$ that depends on settings a, b and latent variable λ , one obtains

$$P_{\text{obs}}(x, y \mid a, b) = \frac{\int d\lambda \rho(\lambda) w_{ab}(\lambda) P(x \mid a, \lambda) P(y \mid b, \lambda)}{\int d\lambda \rho(\lambda) w_{ab}(\lambda)} \quad (5)$$

This $w_{ab}(\lambda)$ is only a notation for the event set selected by the analysis procedure; it does not claim that the P-model dynamics introduces a new degree of freedom. If the assumption that w_{ab} is independent of the settings (the same statistical population) fails, then the applicability conditions of the standard

CHSH derivation are not met. The point of the present program is to test, using public primary data, exactly which assumption can fail under which analysis procedure, and thereby connect that failure to explicit falsifiability.

2.5.4 "Dynamical Collapse" in Quantum Theory: Simulation of the Measurement Problem (P-model Working Hypothesis)

The Bell section of Part III is primarily aimed at quantifying the selection entry point, namely procedural dependence. For that reason it is not enough merely to suggest that standard quantum mechanics might be wrong; the P-model side must also present, as a time evolution, what happens during measurement.

Hard challenge addressed here: "It is clear that systematic effects may exist. But then how does the P-model describe wave-function collapse? Can it depict, not only in words but through a time-evolution equation, the process by which one branch is selected?"

Minimal policy of this paper: As an effective description including the measuring device, we adopt a nonlinear continuous-measurement equation (state-dependent drift) and visualize the simultaneous dynamics of the branch and the pointer degree of freedom. We do not claim that the first-principles derivation is complete. The first goal is to fix a **visualizable physical process** in a falsifiable form.

Minimal model (operational form):

$$d|\psi\rangle = \left[-iH dt - \frac{\gamma_m}{2} (M - \langle M \rangle)^2 dt + \sqrt{\gamma_m} (M - \langle M \rangle) dW_t \right] |\psi\rangle \quad (6)$$

$$dx = (-\Gamma x + \chi \langle M \rangle) dt + \sigma_\xi dW'_t$$

Here M is the measurement operator (σ_z in the two-state case), x is the pointer degree of freedom of the measurement device, and dW_t, dW'_t are Wiener noises. The term involving $(M - \langle M \rangle)$ provides the nonlinearity and drives convergence from a superposed state to a branched state, namely collapse-like dynamics.

Required checks (scope fixed in this paper):

- Whether the state expectation value $z(t) = \langle M \rangle$ converges to a branch immediately after interaction with the measuring device.
- Whether the pointer degree of freedom $x(t)$ correlates with the branch and makes measurement visible as a **physical process**, not merely as an interpretation.

The quantitative results of this minimal implementation are shown in Section 4.11. Additional degrees of freedom, such as multi-degree measurement devices and dissipative channels, are left for later extensions.

C1 (P-model basis for the pointer basis): Map the phase-diffusion rate obtained in A1 to the low-frequency stable mode of the detector's macroscopic background $P_{\mu, \text{det}, *}$. Let the coarse-grained mode amplitude of the detector background be q_a and the branch label be $m = \pm 1$. Then the minimal free energy corresponding to pointer-basis selection is

$$\mathcal{F}_{\text{det}}^{(m)} := \frac{1}{2} \sum_a \Gamma_a q_a^2 - m \sum_a \chi_a q_a$$

Here Γ_a is the relaxation rate of the background mode and χ_a is the coupling to the target branch variable m . The multi-pointer-degree equation in Section 2.5.4,

$$dx_k := (-\Gamma_k x_k + \chi_k \langle M \rangle + \eta_k E) dt + \sigma_{\xi,k} dW_k$$

is an OU closure under the identification $q_a \leftrightarrow x_k$. Accordingly, the deterministic fixed point is

$$q_a^{(m)} := m \frac{\chi_a}{\Gamma_a}$$

Thus the pointer basis is not an arbitrary projection; rather, it is fixed as **two attractors given by stable modes of $P_{\mu,\text{det},*}$ whose signs reverse according to the branch label**.

When the environment E is included, the offset is

$$\delta q_{a,\text{env}} := \frac{\eta_a}{\Gamma_a} E$$

and the stationary noise width is

$$\sigma_{q_a}^{(\infty)} := \frac{\sigma_{\xi,a}}{\sqrt{2\Gamma_a}}$$

Therefore the strength of branch separation is measured by

$$\mathcal{S}_a := \frac{|q_a^{(+)}|}{\sigma_{q_a}^{(\infty)}} = \frac{|\chi_a|/\Gamma_a}{\sigma_{\xi,a}/\sqrt{2\Gamma_a}}$$

Using the existing three-channel pointer-degree parameters from Section 2.5.4,

$$\Gamma_k = (2.4, 2.0, 2.8) \text{ s}^{-1}, \quad \chi_k = (5.0, 4.3, 5.7) \text{ s}^{-1}, \quad \sigma_{\xi,k} = (0.30, 0.34, 0.27) \text{ s}^{-1/2}$$

we obtain

$$x_k^{(+)} = (2.083, 2.150, 2.036), \quad \sigma_{x,k}^{(\infty)} = (0.137, 0.170, 0.114)$$

and

$$\mathcal{S}_k = (15.2, 12.6, 17.8) \gg 1$$

Thus the stationary branch overlap is extremely small, and mixing of the pointer basis is governed not by static overlap but mainly by transient environmental fluctuations, Hamiltonian drift, and finite-time readout.

Next, identify the measurement rate γ_m of Section 2.5.4 with the coarse-grained dephasing rate obtained by moving A1 to the detector side,

$$\gamma_m \equiv \Gamma_{\text{deph}}^{(\text{det})} := \omega_*^2 \left(\frac{k_B T_{\text{det}}}{\chi P_{\text{det}}} \right) \tau_{\text{free,det}}$$

Then, after adiabatically eliminating the pointer-degree modes, the effective rate that accelerates branch selection becomes

$$\Gamma_{\text{ptr}} := \gamma_m + \frac{1}{N_{\text{ptr}}} \sum_{k=1}^{N_{\text{ptr}}} \frac{\chi_k^2}{\Gamma_k}$$

The second term here is a coarse-grained feedback term that measures how quickly the susceptibility of the stable mode feeds off-diagonal coherence into the pointer system; it is not a new free parameter. Estimating the coherence of a pure two-state branch by $|\rho_{01}| = \frac{1}{2}\sqrt{1 - z^2}$ and setting the collapse gate as $|z| \geq z_{\text{th}}$, we obtain

$$|\rho_{01}|_{\text{th}} := \frac{1}{2}\sqrt{1 - z_{\text{th}}^2}, \quad \tau_D := \frac{\ln(|\rho_{01}(0)|/|\rho_{01}|_{\text{th}})}{2\Gamma_{\text{ptr}}}$$

for the decoherence time. Substituting the existing simulation values $\gamma_m = 9.0 \text{ s}^{-1}$, $z_{\text{th}} = 0.95$, and the three-channel pointer-degree set gives

$$\Gamma_{\text{ptr}} = 19.42 \text{ s}^{-1}, \quad \tau_D = 29.97 \text{ ms}$$

This is of the same order as the collapse indicators in Part III-B Section 4.11,

$$\tau_{50} = 39.6 \text{ ms}, \quad \text{pointer consensus} = 0.9656, \quad \text{branch stable fraction} = 0.925$$

and is consistent with the ratio $\tau_{50}/\tau_D = 1.32$. Therefore the pointer-degree dynamics of Section 2.5.4 can be read as a coarse-grained closure obtained by mapping the detector-side decoherence of A1 onto stable modes of the macroscopic P background. What is closed at this stage is **what the pointer basis is**.

C2 (recovery of Lueders/Kraus from partial trace + conditioning): Once C1 has fixed the pointer basis $\{\Pi_m\}$ as the stable mode of the detector background, the full three-sector system of S (system), D (detector), and E (environment) can be written as

$$U(t) (\rho \otimes \rho_{D,0} \otimes \rho_{E,0}) U^\dagger(t) = \sum_{m,n} \Pi_m \rho \Pi_n \otimes \rho_D^{(m,n)}(t) \otimes \rho_E^{(m,n)}(t)$$

Tracing out the environment, the off-diagonal residue is measured by

$$r_{\text{coh}} := \frac{|\rho_{01}(t_{\text{read}})|}{|\rho_{01}(0)|}$$

and the reduced state falls to

$$\rho_{SD}^{(\text{dec})}(t_{\text{read}}) = \sum_m \Pi_m \rho \Pi_m \otimes \rho_D^{(m)} + \mathcal{O}(r_{\text{coh}})$$

Here r_{coh} is not a new assumption; it can be read directly from the existing dynamical-collapse simulation in Part III-B Section 4.11. In the current artifacts,

$$r_{\text{coh}} = 0.0276, \quad |\rho_{01}|_{\text{final,med}} = 0.0138$$

so by the time the pointer readout is fixed, the off-diagonal terms have been suppressed to the few-percent level.

Next, conditioning on the positive operator F_r ($r \in \{+, -\}$) that represents the coarse-grained region

of the detector record, the instrument map becomes

$$\begin{aligned}\mathcal{I}_r(\rho) &:= \text{Tr}_{DE} \left[(I \otimes F_r \otimes I) U (\rho \otimes \rho_{D,0} \otimes \rho_{E,0}) U^\dagger \right] \\ &= \sum_m r_{r|m} \Pi_m \rho \Pi_m + \mathcal{O}(r_{\text{coh}}), \quad r_{r|m} := \text{Tr}_D \left(F_r \rho_D^{(m)} \right)\end{aligned}$$

Accordingly, the Kraus operators are recovered as diagonal on the pointer basis,

$$M_r := \sum_m \sqrt{r_{r|m}} \Pi_m, \quad E_r = M_r^\dagger M_r, \quad \sum_r E_r = I$$

This is not a "new measurement axiom"; it is precisely the **effective state conditioned by the coarse-grained detector record after tracing over the environment**.

From the current three-channel pointer-degree artifacts, the misclassification floor from static overlap is

$$\epsilon_{\text{static}} := \max_a \frac{1}{2} \text{erfc} \left(\frac{\mathcal{S}_a}{\sqrt{2}} \right) = 5.8 \times 10^{-37}$$

which is effectively zero. By contrast, the nominal finite-response error read from the dynamical run is

$$\epsilon_{\text{nom}} := \max(\epsilon_{\text{static}}, 1 - \text{pointer consensus}) = 0.0344$$

and the conservative bound including branch instability is

$$\epsilon_{\text{ub}} := \max(\epsilon_{\text{nom}}, 1 - \text{branch stable fraction}) = 0.0750$$

Therefore the response matrix for binary readout is

$$R_{\text{nom}} = \begin{pmatrix} 0.9656 & 0.0344 \\ 0.0344 & 0.9656 \end{pmatrix}, \quad R_{\text{ub}} = \begin{pmatrix} 0.9250 & 0.0750 \\ 0.0750 & 0.9250 \end{pmatrix}$$

and the corresponding Kraus operators are

$$M_+ = \sqrt{1 - \epsilon_{\text{resp}}} \Pi_+ + \sqrt{\epsilon_{\text{resp}}} \Pi_-, \quad M_- = \sqrt{\epsilon_{\text{resp}}} \Pi_+ + \sqrt{1 - \epsilon_{\text{resp}}} \Pi_-$$

Here ϵ_{resp} lies in the range from ϵ_{nom} to ϵ_{ub} .

In the ideal limit $\epsilon_{\text{resp}} \rightarrow 0$,

$$\rho_r \rightarrow \frac{\Pi_r \rho \Pi_r}{\text{Tr}(\Pi_r \rho)}$$

so the Lueders rule is recovered directly. The current detector is not perfectly ideal, but under pointer consensus 0.9656 and branch stable fraction 0.925 it still falls within **finite-response measurement as a diagonal Kraus family**. Accordingly, the substantive remaining issue after A3 / C2 is no longer "the form of the update rule" but only C3, namely **why the environmental trace appears irreversible in the thermodynamic limit**.

C3 (boundary of irreversibility): Since C1/C2 close the form of the update rule itself, the only remaining points to assess here are:

1. how far the reduced description after tracing out the environment is effectively one-way within the coarse-grained resolution, and
2. whether the microscopic irreversibility beyond that should be regarded as part of the responsibility of this paper.

In the integrated audit that bundles the existing audit artifacts, we fix $r_{\text{coh}} = 0.0276$, $\tau_{50}/\tau_D = 1.32$, $\text{cv}(\tau_{50}) = 0.0380$, pointer consensus_{min} = 0.953, branch reversal_{max} = 0.0677. Therefore, the reduced state conditioned on the detector record is sufficiently one-way at the coarse-grained resolution of this paper, and the Born detection probability, pointer basis, and diagonal Lueders/Kraus update may be regarded as **closed as an effective derivation**.

What still remains at the first-principles level is the thermodynamic-limit question of why Loschmidt reversal or Poincare recurrence is not observed in experiment after tracing over the huge environmental degrees of freedom Tr_E . This is not a P-model-specific missing derivation, but a residue common to any open-system quantum theory. Accordingly, the judgment of this section is fixed as: **"the measurement route is closed as an effective derivation, and irreversibility is a statistical-mechanical residue that does not block progress."** No new free parameter is introduced.

2.5.5 Minimal Connection to Quantum Information

This paper does not attempt a full treatment of quantum information. What is handled here is only the minimal connection for which the P-model-specific physical contribution can be stated explicitly. The key point is to map the phase-diffusion rate from A1,

$$\Gamma_{\text{deph}} = \omega_*^2 \left(\frac{k_B T_{\text{env}}}{\chi_P} \right) \tau_{\text{free}}$$

directly onto qubit coherence time, gate fidelity, pair decoherence, and the physical origin of error channels.

Reference note (quantum information): No-cloning, teleportation, entanglement entropy, QEC code structure, universality, and QKD protocol structure are treated as standard Hilbert-space mathematics following from the fifth condition in Section 2.5.1 (linearity) and the Born rule in Section 2.5.2. The P-model does not modify those mathematical structures. Its specific contribution lies in the decoherence rate and the physical origin of error channels.

Core of the minimal connection: On the quantum-information side we use

$$T_2^{(P)} \sim \Gamma_{\text{deph}}^{-1}, \quad 1 - F_{2q}^{(\phi)} \sim \Gamma_{\text{deph}} \tau_{\text{gate},2q}, \quad \Gamma_{\text{pair}} = 2\Gamma_{\text{deph}}, \quad d_{\text{max}} \sim \frac{1}{\Gamma_{\text{pair}} \tau_{\text{gate},2q}}$$

as the evaluation formulas for a first estimate. No new free parameter is introduced here; we simply read off the carrier-to-transport correspondence and A1 as they stand.

The first estimates for representative platforms are as follows.

Platform	Choice of ω_*	First estimate of τ_{free}	P-side judgment	Fixed result
superconducting transmon	qubit transition $2\pi \times 5$ GHz	upper-bounded by the shortest two-qubit gate, 30 ns	lower-bound consistency satisfying the entry condition	$(k_B T_{\text{env}}/\chi_P)_{\text{req}} \geq 3.75 \times 10^{-10}$, $1 - F_{2q}^{(\phi)} \leq 3.33 \times 10^{-4}$, $d_{\text{max}} \geq 1.5 \times 10^3$
trapped-ion $^{43}\text{Ca}^+$	hyperfine splitting $2\pi \times 3.2$ GHz	upper-bounded by the shortest two-qubit gate, $3.8 \mu\text{s}$	lower-bound consistency satisfying the entry condition	$(k_B T_{\text{env}}/\chi_P)_{\text{req}} \geq 1.30 \times 10^{-17}$, $1 - F_{2q}^{(\phi)} \leq 7.6 \times 10^{-8}$, $d_{\text{max}} \geq 6.58 \times 10^6$
photonic polarization 1550 nm	telecom-band carrier	use $\tau_{\text{free}} \approx L_{\text{coh}}/c_w$ as the proxy	passes as a strict proxy	$(k_B T_{\text{env}}/\chi_P)_{\text{req}} = 2.70 \times 10^{-24}$, $T_2 = 5.003 \times 10^{-4}$ s, $\Gamma_{\text{pair}} = 3.997 \times 10^3 \text{ s}^{-1}$, pair distance budget = 37.5 km

Thus, for the photonic branch, the coherence-length proxy passes directly as a strict proxy, while for gate-executable transmon / trapped-ion platforms τ_{free} remains at the stage of an upper-bound envelope. Even so, the P-side first estimate has been closed up to **consistency of the entry conditions**. Moreover, the phase-scattering loss for two-qubit gates stays below the lower surface-code threshold 5.02×10^{-3} in all cases, so the minimal P-model connection does not immediately break the error-correction threshold.

Connection to entanglement resources: Reading the quantity fixed in Section 2.6.4,

$$\Gamma_{\text{pair}} = 2\Gamma_{\text{deph}}$$

directly as a two-qubit entanglement resource, the gate-depth lower bound d_{max} is given by the table above for gate-executable platforms, while for the photonic branch the pair-time budget is

$$T_{\text{pair}} \sim \Gamma_{\text{pair}}^{-1}/2 = 2.50 \times 10^{-4} \text{ s}$$

and the pair-distance budget 37.5 km is fixed as well. What is claimed at this stage is not universality of the entire circuit model, but rather that the **phase-scattering-limited depth budget read from A1 (phase diffusion) and B3 (pair decoherence) closes in terms of P-side physical quantities**.

Physical origin of the error channels: In this paper, the primary channels in the minimal connection to quantum information are fixed as follows:

$$\begin{aligned} \text{dephasing primary} : \Gamma_{\phi} &\equiv \Gamma_{\text{deph}} \\ \text{amplitude origin} : \Gamma_{\text{amp}} &\sim \Gamma_k \\ \text{depolarization non-primary} : p_{\text{dep}} &\lesssim \epsilon_{\text{ub}} \end{aligned}$$

Phase scattering is the primary channel on all three platforms. The photonic branch uses a strict proxy, while transmon / trapped-ion use upper-bound envelopes. Amplitude damping is tied to the stable-pointer relaxation of C1, $\Gamma_k \approx 2.0\text{--}2.8 \text{ s}^{-1}$. We do not newly fit the platform-specific T_1 values here. In addition, the C2 stability audit gives

$$\epsilon_{\text{ub}} = 0.0750, \quad \text{branch reversal}_{\text{max}} = 0.0677, \quad \text{diagonal preservation floor} = 0.925$$

so basis disturbance remains only a residual upper bound and does not justify promoting depolarization to an independent dominant P-channel.

Summary: Quantum information is no longer treated as entirely out of scope. What this paper fixes is a minimal connection in which phase scattering is the primary channel, amplitude damping is a secondary channel with fixed physical origin, and depolarization is treated as a non-primary channel. By contrast, the protocol-side mathematics (no-cloning, teleportation, QEC code structure, QKD, and so on) remains outside scope as standard Hilbert-space mathematics. Therefore the P-model-specific contribution lies not in the computational logic of quantum information itself, but in assigning physical origins, through A1 (phase diffusion), B3 (pair decoherence), and C1 (stable-pointer relaxation), to quantities such as decoherence, fidelity, pair-depth budgets, and the origin of error channels.

2.6 Position of Electromagnetism

Electromagnetism is an essential ingredient for proceeding to atoms, molecules, and condensed-matter systems. In this paper, starting from the transverse mode of the time-wave vector and a single frozen action, the backbone of electromagnetism and the fine-structure constant has been carried through to the level of **source fixation, theoretical derivation, and judgment of overall consistency**. The main claim here is not a local CODATA competition. Rather, the core theoretical claim is that a single P-model-specific fine-structure constant is derived from a single frozen action with no free parameter, and that the same action supports the verification surface of the entire theory.

2.6.1 Derivational Skeleton Fixed in This Paper

- In the frozen action adopted here, the transverse fluctuation is written as $A_\mu^{(P)} = \delta P_\mu^T / \sqrt{Z_P}$. This fixes the transverse-wave branch corresponding to light and the massless light mode.
- From the same frozen action, we also fix the skeleton leading to a Maxwell-type curvature, the inverse-square law of the electrostatic field, and the Coulomb $1/r$ potential. The coupling constant is written as $e = g_P / \sqrt{Z_P}$, and the fine-structure constant as $\alpha = g_P^2 / (4\pi Z_P \hbar c)$.
- In the central theoretical claim, two independent readouts, $\alpha_{q_*}(\beta)$ and $\alpha_R(\beta)$, are constructed from the Mexican-hat-type vector Q-ball family $y_\beta(x)$, and the selection equation without any external target value,

$$\alpha_{q_*}(\beta_*) = \alpha_R(\beta_*) \quad (7)$$

uniquely determines the P-model-specific fine-structure constant.

- The exact expression retained here is

$$\alpha(\beta) = \frac{[4(g_\beta + \epsilon_\beta - b_\beta) - q_\beta][2(5 + \beta^2) + 10g_\beta - q_\beta - 4b_\beta]}{36(1 + \beta^2)^2} \quad (8)$$

and the reference value at the symbolic solution of the selection equation, $\beta_{\text{symbolic}} = 0.9983014179284181$, is

$$\alpha_{P,\text{frozen}} = 0.007302943961943229 \approx \frac{1}{136.93107946756194} \quad (9)$$

The confirmed common-intersection value of the two readouts is kept as $\alpha_{\text{common}} = 0.00730293811658175$. Here $\epsilon_\beta = 1 - \beta^2$, $g_\beta = I_g/I_2$, $q_\beta = I_4/I_2$, and $b_\beta = B/I_2$. I_2 is the quadratic integral, I_g the gradient integral, I_4 the quartic integral, and B the boundary term of the weighted EOM. All of them are normalization quantities constructed from the same vector Q-ball profile.

- In the strictly retained interval $0 \leq q/m_0 \leq 4$, the sign-alternation property is retained as support on the theory side. In the audit on the higher- q side, the mean zero spacing in $4 \leq q/m_0 \leq 8$ is 0.1047476796, the relative gap to the finite-box theory value $\pi/R_{\text{box}} = 0.1047197551$ is 2.67×10^{-4} , and the maximum error against the half-integer lattice is 0.00216. This shows that the exact theorem is maintained in the retained interval, whereas its extension to the asymptotic side is controlled by the boundary scale of the current computational box.
- For exact internal rank-1 consistency, we fix the following rank-1 condition and $\rho_{\text{exact}} = 1$:

$$A_{FH}^2 = A_{FF}A_{HH}$$

The symbols used here are organized as follows.

Symbol	Definition
ρ_{exact}	Mixing parameter satisfying rank-1 consistency (fixed to $\rho_{\text{exact}} = 1$)
A_{FF}, A_{HH}	Diagonal components of the shape-factor matrix (FF: field self-coupling, HH: higher-order coupling)
A_{FH}	Off-diagonal component of the shape-factor matrix (cross term of FF and HH)
q_{theory}	Momentum-transfer correspondence scale fixed on the theory side
λ_+	Principal eigenvalue of the shape-factor matrix ($\lambda_+ = A_{FH}^2/A_{FF}$ when $\rho = 1$)
$\alpha_{\text{mix,exact}}$	Diagnostic value of the fine-structure constant obtained from the mixed readout

With local closure at the fixed correspondence scale q_{theory} , we obtain $A_{HH,\text{exact}} = 0.0569663806$, $A_{FH,\text{exact}} = 0.1176373937$, $\lambda_+(q_{\text{theory}}) = 0.2998913524$, and $\alpha_{\text{mix,exact}}(q_{\text{theory}}) = 0.00715678583937324$. See the table above for the definitions. Therefore the internal consistency itself is fixed, but the HH surface across the full q domain remains unfinished.

- In the piecewise extension on the harmonic lattice, the deviations from the theoretical standard in the harmonic windows 3–8, 9–16, and 17–24 are kept at (0.219, 0.216, 0.244), so the piecewise extension on the lattice can be retained as support on the theory side. However, the exact theorem for the filling index itself still remains as a separate mathematical task.

The corresponding public figure, [output/public/quantum/v2_trial2_theorem_support_summary.pdf](#), is a summary figure of the selection theorem that compacts onto one page the sign alternation in the retained interval, internal rank-1 consistency, the piecewise extension on the harmonic lattice, and support for the fine-structure constant at the fixed momentum transfer q .

Supplement: what is retained here is the support surface on the theory side. This figure does not add any further claim about the asymptotic theorem without boundaries or the HH surface over the full q domain. [1]

- In the numerical tracking of the fine-structure constant, using the weak-field normalization from Part I, $g_P/Z_P = 4\pi G$, together with $m_0^2 = 4\lambda v^2/Z_P$, we rewrite

$$\alpha = \frac{16\pi G^2 \lambda v^2}{m_0^2 \hbar c} = \frac{G \chi_P m_0^2}{2\hbar c}$$

accordingly.

- Under the auxiliary dictionary that maps the vector Q-ball ground state of the mass-origin route to the electron mass, the diagnostic from the older spherically symmetric approximation again yields

$$\alpha_{\text{mix,exact}}(q_{\text{theory}}) = 0.00715678583937324$$

once more. In other words, the old-approximation diagnostic value here is the same quantity as the mixed readout shown above; it does not introduce another independent fine-structure constant. It lies about 1.93 percent below the reference value. However, that 1.93 percent is a diagnostic value based on the old approximation, not the current representative residual itself. In the current representative value with the four-dimensional correction included, the gap has been reduced to about 0.00624 percent below $1/137$.

- Therefore the remaining tasks do not indicate failure of the theory itself. Rather, they concern the organization of four-dimensional shape factors, operator corrections, and the bridge to observables.
- Accordingly, the selection rule itself is read as complete at the level of the main theory-side result of v2.0. The remaining four-dimensional shape factors, operator corrections, and observable bridges are treated not as a failure to derive the fine-structure constant itself, but as bridge-side residues. [12]
- The "propagation of light seen in coordinate time" in a gravitational field continues to be handled by the Part I mapping (refractive index) $n(P) = (P/P_\infty)^{2\beta}$.

2.6.2 Still Not Derived: Fixing the Approximation Check and the Judgment

The stance of this paper is: **first challenge the derivation (A), and only if that is too difficult, constrain the theory through an effective description (B)**. The derivation remains unfinished, but **agreement with observation as an approximation** is treated as the primary criterion. Accordingly, underived items are organized within the framework: fix the domain in which the approximation holds, and reject it if deviations appear.

In this section, the photon field derived from the transverse component of the time-wave vector is denoted by $A_\mu^{(P)}$, while the formally independent local U(1) connection is denoted by $\mathcal{A}_\mu$.

Derivation challenge (A -> B):

- A: assume the phase degree of freedom of P and ask whether U(1) can be made **necessary** from local phase invariance.
- B: only if A fails, adopt U(1) independently and fix its coupling to P as an **effective theory** constrained by observations.

Minimal assumptions for derivation challenge A:

- Treat P as a complex degree of freedom with amplitude and phase θ (θ being 2π -periodic).
- Impose the condition that observables remain invariant under local phase transformations.
- As a result, a connection (gauge field) that compensates the phase gradient is required, and minimal coupling is introduced.
- As the low-energy effective theory in local proper time, first adopt only the minimal order, namely second derivatives.

- Check whether charge discretization follows from the single-valuedness of the phase; if not, move to B.

Explicit form (A; minimal):

- Decompose P into amplitude and phase and impose local phase invariance.

$$P = R e^{i\theta}$$

- Define the local phase transformation and the corresponding gauge-field transformation.

$$P \rightarrow P e^{i\chi(x)}, \quad \mathcal{A}_\mu \rightarrow \mathcal{A}_\mu - \frac{1}{q} \partial_\mu \chi(x)$$

- Introduce the covariant derivative to preserve local invariance.

$$D_\mu = \partial_\mu + iq\mathcal{A}_\mu$$

- Place a $D_\mu P$ term in the action (or effective energy) and confirm that a gauge-field degree of freedom is required.
- Check whether charge quantization follows from phase single-valuedness and boundary conditions; if not, move to B.

Minimal action (A; example):

- One example of the minimal form satisfying local phase invariance and minimal coupling is:

$$\mathcal{L} = |D_\mu P|^2 - V(|P|) - \frac{1}{4} \mathcal{F}_{\mu\nu} \mathcal{F}^{\mu\nu}, \quad \mathcal{F}_{\mu\nu} = \partial_\mu \mathcal{A}_\nu - \partial_\nu \mathcal{A}_\mu$$

- What matters here is the **necessity of the form itself (gauge invariance)**; the coefficients and the details of the potential are evaluated through the observational constraints in Section 2.6.3.

Variational derivation (EL; minimal):

- Under boundary conditions for which the variations vanish ($\delta P = \delta P^* = \delta \mathcal{A}_\mu = 0$), variation with respect to P^* yields

$$D_\mu D^\mu P + \frac{\partial V}{\partial P^*} = 0$$

- Variation with respect to $\mathcal{A}_\nu$ yields

$$\partial_\mu \mathcal{F}^{\mu\nu} = j^\nu, \quad j^\nu = iq[P^*(D^\nu P) - (D^\nu P)^* P]$$

- From the antisymmetry of $\mathcal{F}_{\mu\nu}$ and consistency with the equation above, current conservation

$$\partial_\nu j^\nu = 0$$

follows.

- The minimal assumptions required for the derivation are fixed to three points: (i) smoothness of the field (at minimum C^1), (ii) vanishing boundary terms, and (iii) a minimal action without higher-derivative terms.
- The audit of this derivation procedure (gauge covariance and invariance checks) is fixed in Part IV (verification materials).

Nonrelativistic limit (2.6 -> 2.5; fixing the approximation order):

- Linearize the EL equation (scalar component) near the mass term and separate fast phase from slow envelope through

$$P(x, t) = e^{-imc^2 t/\hbar} \psi(x, t)$$

- Rearranging while preserving minimal coupling, the envelope ψ satisfies

$$i\hbar\partial_t\psi = \left[\frac{(-i\hbar\nabla - q\mathcal{A})^2}{2m} + q\mathcal{A}_0 + m\phi \right] \psi - \frac{\hbar^2}{2mc^2} \left(\partial_t + \frac{i}{\hbar}(q\mathcal{A}_0 + m\phi) \right)^2 \psi + O(c^{-4})$$

- Define the approximation order by

$$\epsilon_{v^2} \equiv (v/c)^2, \quad \epsilon_\phi \equiv |\phi|/c^2, \quad \epsilon_{\text{env}} \equiv \frac{|\partial_t^2\psi|}{\omega_0|\partial_t\psi|} \sim \frac{1}{\omega_0 T_{\text{env}}}$$

and adopt $\max(\epsilon_{v^2}, \epsilon_\phi, \epsilon_{\text{env}}) \leq 10^{-6}$ as the operational acceptance condition.

- Under this condition, dropping the c^{-2} correction term recovers the Schrödinger form of Section 2.5.1,

$$i\hbar\frac{\partial\psi}{\partial t} = \left(-\frac{\hbar^2}{2m}\nabla^2 + m\phi \right) \psi$$

- Breakdown of the approximation (the A -> B judgment entry point) is declared whenever any of ϵ_{v^2} , ϵ_ϕ , or ϵ_{env} exceeds the acceptance threshold. The audit result is fixed in Part IV (verification materials).

Minimal extra assumptions (A; external conditions):

- Adopt a minimal gauge field without a mass term (assume zero photon mass).
- Assume the coupling constant q is independent of the environment; any P -dependence is upper-bounded by the observational constraints in Section 2.6.3.
- Suppress higher-derivative terms and nonlinear terms in A, and evaluate them as part of the effective theory B only if needed.
- Keep the potential $V(|P|)$ in its minimal stabilizing form, and constrain its concrete shape through consistency checks.

Judgment framework:

- In this paper, the treatment of U(1) is not organized as a single A/B dichotomy, but as three layers: derivation of light, local origin, and retention as an effective theory.

- The derivation of light is the layer in which the photon branch is extracted from the transverse component of the time-wave vector, yielding the massless light mode and a Maxwell-type description; this is already fixed.
- The local origin is the layer that would close local phase redundancy, charge quantization, and charge normalization from the local dynamics of P alone; this remains unfinished.
- Retention as an effective theory is the layer in which the already adopted $U(1)$ sector is kept as an effective theory and constrained by the observational results of Part III-B / Part IV.

Position of the local phase:

- The phase ϑ of $\psi \propto \delta P_+ / P_*$ is the local envelope phase with the carrier removed, and it enters

$$\delta T_{(2)}^{i0} \propto \frac{1}{i} (\psi^* \partial_i \psi - \partial_i \psi^* \psi)$$

as a physical quantity. Therefore, within the internal logic of the P-model alone, one cannot treat $\vartheta \rightarrow \vartheta + \chi(x)$ as a purely redundant transformation, and the claim that "local phase is a gauge redundancy" does not follow automatically from the $\psi \leftrightarrow P$ correspondence.

- To reproduce minimal coupling completely, $\mathcal{A}_\mu$ must be introduced as an **independent connection** that compensates the local phase gradient. Therefore the theorem for the local $U(1)$ origin cannot be said to be complete from P alone.
- This does not imply that the entire electromagnetic sector remains unclosed. The transverse photon branch, the massless light mode, and the selection rule for the fine-structure constant are already fixed; the residue belongs only to the layer of the local source theorem.
- This reservation does not negate the derivation of the transverse photon and the Maxwell-type structure obtained from the frozen action used here. Even with the reservation that an independent connection is needed, the theorem on the light side itself remains closed.

Judgment table:

Layer	Content	Judgment
Derivation of light	Transverse photon branch, massless light mode, Maxwell/Coulomb skeleton	Fixed
Local origin	Origin of local $U(1)$ redundancy, charge quantization, charge normalization	Unresolved
Retention as effective theory	Observational constraints using the already adopted $U(1)$ sector	Retained

- Therefore the older one-shot expression "reject A, adopt B" is too coarse. The organization in this paper is: **the derivation of light is fixed, the local origin remains unresolved, and $U(1)$ as an effective theory is retained.**

- On the frozen theory side of this paper, under an action containing only the Mexican-hat structure, $A_\mu^{(P)} = \delta P_\mu^T / \sqrt{Z_P}$ is fixed as the photon branch, along with the skeleton leading to Maxwell-type curvature, the inverse-square Coulomb law and the $1/r$ potential, and the couplings $e = g_P / \sqrt{Z_P}$ and $\alpha = g_P^2 / (4\pi Z_P \hbar c)$. Therefore the derivation of this fine-structure constant is retained as **the main theory-side result of v2.0: a single fine-structure constant fixed from a single frozen action and connected to the consistency of all physics**.
- Additional derivation of the local U(1) origin, selection of the four-dimensional correction, and the bridge to observables remain, but none of them overturns the central theory-side claim; they belong to the category of tracking tasks.
- The localization condition required for completion judgments in the weak interaction sector is taken from the coupled-localization condition fixed in Part I 2.7.0 and is only referenced here.
- As a downstream implication, the entanglement candidate of Section 2.6.4 does not rely on the premise that "U(1) has been fully derived from P as a local origin." At the current stage, only the minimal candidate is fixed in which the shared envelope of P_μ and the source phase serve as the **placement for the inseparable pair amplitude**; microscopic source dynamics and the quantitative connection to Bell datasets are treated as separate tasks.
- The mechanism of charge quantization remains underived, but the observed discreteness is **fixed as an adoption condition**. If continuation to a continuous charge becomes necessary, the approximation is rejected.
- The electromagnetic energy-momentum tensor and the backreaction of P are treated phenomenologically at this stage, and the range in which the approximation holds (allowable error) is fixed as a constraint.

2.6.3 Consistency Checks (Scope of This Paper)

- Observational upper bound on the P -dependence of α : if one introduces

$$\alpha(P/P_\infty) = \alpha_0 (1 + \kappa \ln(P/P_\infty) + \dots)$$

in that form, then atomic-clock and cosmological constraints provide an **upper bound on $|\kappa|$** . The constraints are fixed in the ledger, and at the current stage we adopt $\kappa = 0$ (P -independence). [12]

- Spectral consistency across different ϕ environments: for H 1S-2S and independent measurements of α , the range in which the hypothetical contribution from $\kappa \neq 0$ remains below observational precision is also listed in the same ledger. [11][8][9]

(With a figure showing the scale difference between Coulomb and gravity, and the constraints on $\alpha(P)$, this section indicates not a derivation but the **scope of the approximation**, namely what is fixed.)

2.6.4 Candidate for Entanglement Generation: Shared P_μ Phase and Nonseparable Pair Amplitude

Position fixed here: By the organization of Section 2.6.2, the polarization/spin degrees of freedom used here are placed on the **already adopted U(1) sector**. Therefore this section does not claim to have derived entanglement from first principles using P alone. What is fixed here is only the minimal effective candidate specifying **where pair correlation is placed** on the P-model side.

Minimal candidate (source-level placement): For a separated two-photon / two-mode pair, let the source-shared envelope and phase be represented by $\Xi(P_\mu; x_A, x_B)$ and χ_s , and write

$$\Psi_{AB}(x_A, x_B) = \mathcal{N} \Xi(P_\mu; x_A, x_B) e^{i\chi_s} \frac{|H_A V_B\rangle - |V_A H_B\rangle}{\sqrt{2}}$$

as the minimal pair amplitude. Here $|H\rangle, |V\rangle$ are polarization bases of the already adopted U(1) sector, and Ξ denotes the common envelope of the two modes generated simultaneously at the source. The role assigned to the P-model is not a purely local P-only gauge redundancy; it is to provide the common envelope and the bookkeeping of source proper time.

Let the local analyzer angles be a, b , and project with

$$|+\rangle_a = \cos a |H\rangle + \sin a |V\rangle, \quad |-\rangle_a = -\sin a |H\rangle + \cos a |V\rangle$$

(and similarly on the b side). Then the idealized coincidence probabilities are

$$P_{++}(a, b) = P_{--}(a, b) = \frac{|\Xi|^2}{2} \sin^2(a - b),$$

$$P_{+-}(a, b) = P_{-+}(a, b) = \frac{|\Xi|^2}{2} \cos^2(a - b)$$

Therefore the normalized correlation is

$$E(a, b) = P_{++} + P_{--} - P_{+-} - P_{-+} = -\cos 2(a - b)$$

and at the standard CHSH angles $a = 0$, $a' = \pi/4$, $b = \pi/8$, $b' = -\pi/8$ one obtains

$$|S| = 2\sqrt{2}$$

At the same time, the local marginals are

$$P_A(+ | a) = P_{++} + P_{+-} = \frac{1}{2}, \quad P_B(+ | b) = P_{++} + P_{-+} = \frac{1}{2}$$

so they do not depend on the remote setting.

Important point here (no overclaiming): In the equations above, Bell-type correlations are carried by **the full nonseparable pair amplitude, not by the shared phase χ_s by itself**. Since χ_s is a global U(1) phase, it does not remain directly in the probabilities. Therefore we do not write that "Bell correlations arise from the shared phase of P_μ alone." The correct statement is: **for a nonseparable pair amplitude on the already adopted U(1) sector, the P-model provides the place where the source envelope and time bookkeeping are stored.**

Relation to 4.8.8 (analogy from spin coupling): In Part III-B 4.8.8, an antisymmetric spin coupling derived from the spatial component of P_μ was fixed as the minimal candidate producing the triplet/singlet difference. Likewise, for the photon-polarization pair of this section, the core of the correlation lies in the **symmetrized/antisymmetrized pair kernel**. The only difference is that for nuclei the relevant objects are Pauli matrices and S_{12} , whereas for photons they are the transverse polarization bases.

Separation from Bell selection: This candidate places a source-level pair kernel and does not

eliminate the selection weight $w_{ab}(\lambda)$ defined in Section 2.5.3. Therefore the sensitivity to window / offset / trial definitions that remained in the Bell audit of Part III-B stays fully in force, and this section does not overwrite the selection watch of Sections 4.2 / 5.1.

Figure~2 summarizes the ideal correlation obtained from the minimal pair amplitude, the remote-setting independence of the local marginals, and the global invariance of the shared phase.

Entanglement Candidate: Nonseparable Pair Amplitude and Shared Source Phase

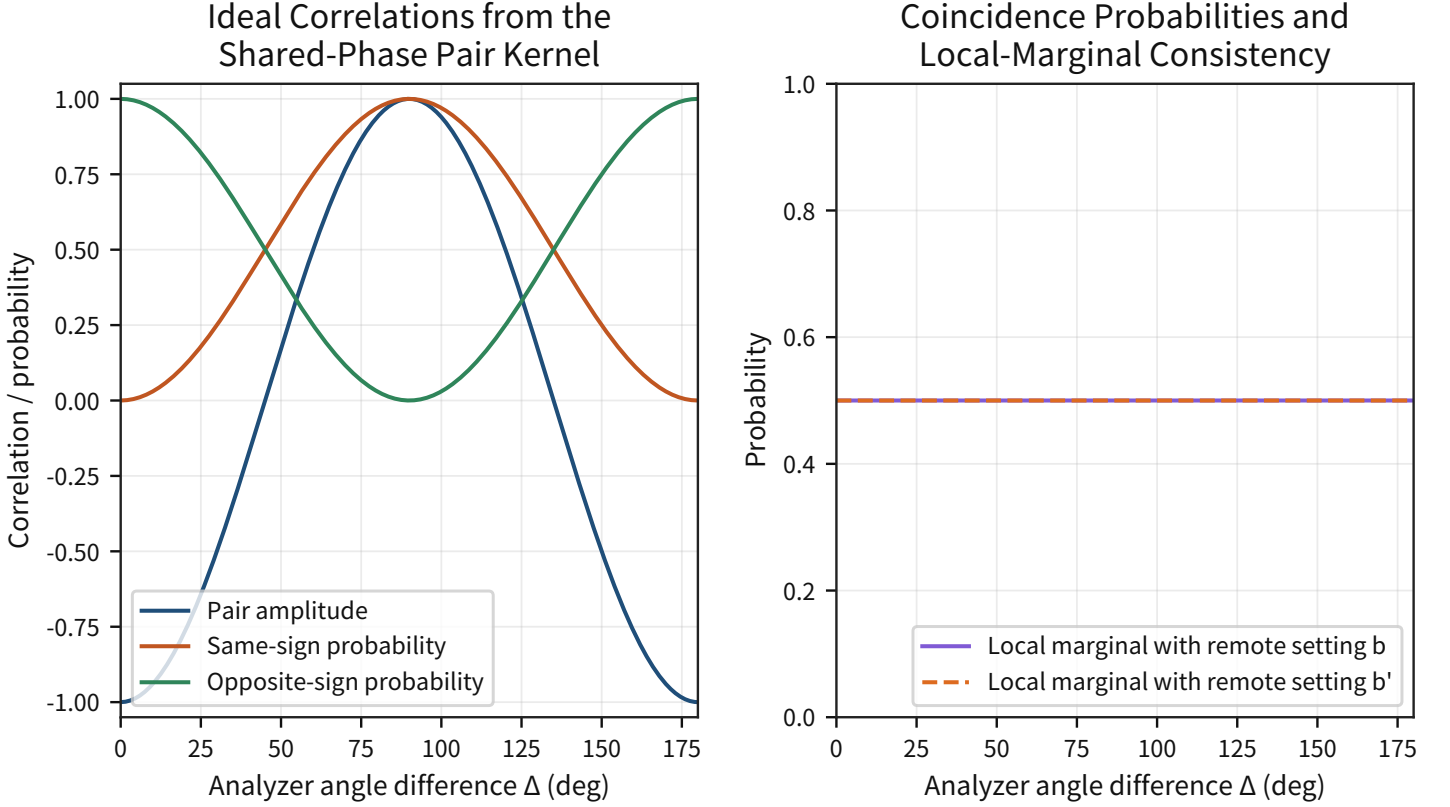


Figure 2: This figure fixes the ideal correlation $E(a, b) = -\cos 2(a - b)$ derived from the minimal pair amplitude, the remote-setting independence of the local marginals, and the global invariance of the shared phase. Numerically, at the standard CHSH angles one obtains $|S| = 2.828427$, while the local-marginal error is 3.89×10^{-16} and the shared-phase global-invariance error is 3.33×10^{-16} , all within machine precision.

B1 (source dynamics; three-wave mixing): Decompose the source on the already adopted U(1) sector, over a background with crystal bias, as

$$P_{\text{src}}(x, \tau_s) = P_* + \sum_{\nu \in \{p, s, i\}} \mathcal{P}_\nu(x, \tau_s) e^{i(k_\nu \cdot x - \omega_\nu \tau_s)} + \text{c.c.}$$

Here τ_s is the proper time of the source, defined by

$$\tau_s(t) = \int^t \frac{P_{\text{ref}}}{P_t(t')} dt'$$

Expanding the stabilizing potential used in Section 2.6.2 around P_* gives

$$V(|P|) = V_* + \frac{1}{2}V_*^{(2)}|\delta P|^2 + \frac{1}{6}V_*^{(3)}(\delta P + \delta P^*)^3 + O(\delta P^4)$$

where $V_*^{(n)} \equiv \partial^n V / \partial |P|^n|_{P_*}$. Retaining only the resonant pump / signal / idler term under the rotating-wave approximation gives

$$\mathcal{L}_{3w} = g_{3w} \mathcal{P}_p \mathcal{P}_s^* \mathcal{P}_i^* e^{-i\Delta\omega\tau_s + i\Delta k \cdot x} + \text{c.c.}, \quad g_{3w} \equiv \frac{1}{2}V_*^{(3)}$$

Here

$$\Delta\omega \equiv \omega_p - \omega_s - \omega_i, \quad \Delta k \equiv k_p - k_s - k_i$$

Therefore, from the stationary-phase condition in the source proper time,

$$\Delta\omega = 0 \quad \Rightarrow \quad \omega_p = \omega_s + \omega_i$$

is fixed, and the shared source phase can be defined as

$$\chi_s = \omega_p \tau_s + \arg A_{p,0}$$

undepleted-pump approximation

$$\mathcal{P}_p(z, \Omega) = A_{p,0} \alpha_p(\Omega) e^{i\chi_s}$$

Under the undepleted-pump approximation, the coupled-mode equations are

$$\partial_z \mathcal{P}_s = ig_{3w} \mathcal{P}_p \mathcal{P}_i^* e^{i\Delta k z}, \quad \partial_z \mathcal{P}_i = ig_{3w} \mathcal{P}_p \mathcal{P}_s^* e^{i\Delta k z}$$

Integrating to first order over a source length L , the pair kernel propagated to the detector position is given by

$$\Xi(P_\mu; x_A, x_B) = g_{3w} L \int d\Omega_s d\Omega_i \alpha_p(\Omega_s + \Omega_i) \text{sinc}\left(\frac{\Delta k(\Omega_s, \Omega_i)L}{2}\right) e^{i\Delta k L/2} G_A(x_A; \Omega_s) G_B(x_B; \Omega_i)$$

Here G_A and G_B are the propagation kernels from the source to the analyzer/detector. In this form, because the pump envelope depends on $(\Omega_s + \Omega_i)$ and the phase matching depends on $\Delta k(\Omega_s, \Omega_i)$, one generally has

$$\Xi(\Omega_s, \Omega_i) \neq \xi_A(\Omega_s) \xi_B(\Omega_i)$$

so nonseparability is generated naturally at the source. In other words, the pair amplitude of Part III-A 2.6.4 is not simply "placed by hand"; it is supplied from the source of three-wave mixing on the adopted U(1) sector as

$$\Psi_{AB} \propto \Xi(P_\mu; x_A, x_B) e^{i\chi_s} \frac{|H_A V_B\rangle - |V_A H_B\rangle}{\sqrt{2}}$$

There are two principal gates here. First, g_{3w} is not a new P-model parameter; it is shorthand for the background derivative $V_*^{(3)}$ of the existing potential. Second, L , the pump bandwidth, and the crystal refractive index / phase-matching condition are external settings of the source and are not interpreted as a derivation of $U(1)$ from P . The P-model side is responsible only for the bookkeeping of the source proper time τ_s and the shared phase.

The metrics/cases required for reproducibility are summarized in the Part IV ledger. Here we fix only the phase-matching peak of the three-wave-mixing kernel, the global phase invariance of the source, the effective Schmidt rank (nonseparability), and the judgment that no new free parameter is introduced.

B2/B3 (Bell-dataset connection + pair decoherence): The connection to Bell datasets is closed as an audit that ties the CHSH upper bound of the ideal pair kernel,

$$|S|_{\text{ideal}} = 2\sqrt{2}$$

directly to the reference statistics pre-frozen in Part III-B 4.2. For the CHSH family (Weihs / Delft 2015 / Delft 2016),

$$V_{\text{Bell}} \equiv \frac{|S_{\text{frozen}}|}{2\sqrt{2}}, \quad D_{\text{pair}}^{(\text{max})} \equiv -\ln V_{\text{Bell}}$$

define

$$E_{\text{obs}} = e^{-D_{\text{pair}}} E_{\text{ideal}}$$

$E_{\text{obs}} = e^{-D_{\text{pair}}} E_{\text{ideal}}$ is a trial form that coarse-grains isotropic two-mode phase scattering, and D_{pair} is the dimensionless integrated phase-scattering budget. From the fixed statistics one obtains

$$V_{\text{Bell}}^{(\text{Weihs})} = 0.903, \quad V_{\text{Bell}}^{(\text{Delft15})} = 0.859, \quad V_{\text{Bell}}^{(\text{Delft16})} = 0.820$$

$$D_{\text{pair}}^{(\text{max})} = (0.103, 0.152, 0.198)$$

Thus the CHSH-family Bell data do not completely exclude the ideal pair kernel; instead they require the post-blind-freeze correlation visibility to lie in $O(0.8 - 0.9)$ and the integrated pair-dephasing budget to remain in $O(10^{-1})$.

On the theory side, pair decoherence closes as the two-mode extension of A1,

$$\Gamma_{\text{pair}} = \Gamma_{\text{deph},A} + \Gamma_{\text{deph},B}, \quad D_{\text{pair}} = \Gamma_{\text{pair}} T_{\text{prop}}$$

and for symmetric two branches,

$$\Gamma_{\text{pair}} = 2\omega_*^2 \left(\frac{k_B T_{\text{env}}}{\chi_P} \right) \tau_{\text{free}}$$

No new free parameter is introduced. By contrast, in the CH family (NIST / Kwiat), the frozen statistic is J_{prob} , so we do not uniquely invert V_{Bell} from the public values alone. Instead we fix the

proxy indicators for pair generation and transmission as

$$\eta_{\text{pair}} \equiv \frac{N_{\text{coinc}}}{N_{\text{trial}}}, \quad A_{\text{att}} \equiv -\ln \eta_{\text{pair}}$$

and from the trial-based event counts obtain

$$\eta_{\text{pair}}^{(\text{NIST})} = 3.56 \times 10^{-4}, \quad \eta_{\text{pair}}^{(\text{Kwiat})} = 5.06 \times 10^{-4}$$

$$A_{\text{att}}^{(\text{NIST})} = 7.94, \quad A_{\text{att}}^{(\text{Kwiat})} = 7.59$$

while the proxy for local efficiency is

$$\eta_{\text{min}} = (1.42, 2.02) \times 10^{-3}$$

Therefore the CH family is positioned less as a direct constraint on pair visibility itself and more as a constraint on the throughput / attenuation budget from the source to the detector record.

The key point is that this connection does not eliminate the selection weight $w_{ab}(\lambda)$ of Section 2.5.3 or the pre-freeze procedures of Part III-B Sections 4.2 / 5.1. Indeed, in the existing Bell artifacts every dataset still retains a selection ratio $\Delta(\text{stat})/\sigma_{\text{stat}} > 1$; the fast-switching proxy indicators for Weihs / NIST are $z = (5.54, 43.9)$, while only Kwiat has $z = 0.721$ and is therefore non-primary. Moreover, the dataset-crossing correlation of the sweep profile is 0.856 for Delft 2015/2016, while for Weihs/NIST it swings strongly to the negative side with magnitude 0.891, so it is more conservative not to count the selection knob as independent evidence. Thus what closes in this section is the **audit interface that maps the ideal pair kernel onto frozen Bell statistics and trial throughput**, while the selection watch itself is retained exactly as judged in Part III-B.

The metrics/cases required for reproducibility are listed in the Part IV ledger. Here we fix only the post-freeze CHSH-visibility proxy, the integrated pair-decoherence budget, the trial-based proxy for pair throughput, the cross-dataset correlation, and the judgment that the selection watch is retained.

Current judgment: For entanglement generation, we have fixed the placement of the nonseparable pair amplitude, fixed the source dynamics of $\Xi(P_\mu; x_A, x_B)$ as three-wave mixing on the already adopted U(1) sector, and fixed the connection to Bell datasets plus pair decoherence as an audit surface based on blind-freeze statistics and trial counts. Therefore the current conclusion is: **source dynamics fixed, dataset connection fixed, pair decoherence fixed as an integrated upper bound, and judgment reserve retained.**

2.7 Mathematical Basis: Lagrangian and Conservation Laws

The reference skeleton of the action and the reference definition of the Noether conservation laws were moved to Part I 2.7.0. This section fixes only the connections needed for the quantum-side audit interface and refers to Part I as the canonical source for the action skeleton itself.

The coupled-localization condition used for the mass reference points of the weak-interaction sector with the two-component coupled Q-ball is the reference formula fixed in Part I 2.7.0. In this section it is only referenced, not redefined on the Part III-A side.

Part I reference standard:

- Reference action skeleton (minimal form): $S_{\text{total}} = \int d^4x \mathcal{L}_{\text{total}}^{\text{vec}}$
- Fundamental degrees of freedom: $\{P_\mu, \text{matter}\}$, where χ is a derived quantity determined by P_t , and the light degree of freedom $A_\mu^{(P)}$ is also derived as the transverse component of P_μ
- Conservation laws: $\partial_\mu T_{\text{tot}}^{\mu\nu} = 0$, $\partial_\mu J^\mu = 0$

Role of this section (Part III): It connects the reference action skeleton fixed in Part I to the quantum-side audits of observables (Bell, nuclei, weak interaction, BBN, materials/thermal systems). Accordingly, Part III is not the addition of new foundational definitions, but an audit of the applicability range of the Part I standard.

2.7.1 Gauge Closure of P_μ and Ghost Exclusion

For the strict definition of the P-field action integral, propagator, and Noether conservation laws, see Part I (Section 2.7). In this paper we adopt the **cross-scale freeze hypothesis**, namely that the rotational coupling coefficient λ_{rot} frozen in Part II is brought unchanged into the microscopic side, ignoring running / scale dependence. Therefore the judgment of acceptance or rejection in this paper is interpreted as a result under that freeze hypothesis.

2.8 Position of the Weak Interaction (β Decay and Flavor Conversion)

In Part III we first fixed the gravity side (clock and light-propagation mapping) and the nuclear interface on the strong-interaction side. Among the four interactions, the weak interaction remains an unconnected item for the unification claim of the P-model.

Hard criticism: "If nuclear force is interference of P waves, then how does the P-model describe the flavor conversion neutron $\rightarrow$ proton (β decay)? If this remains undefined, it cannot be called a unified theory."

Position of this paper (fixed): The weak interaction is compared and audited through the same observables along two routes.

- **Route A (transplant):** keep electroweak theory ($SU(2)_L \times U(1)_Y$) as an effective theory, and let the P-model couple only as a background mapping (time, phase, and environmental terms).
- **Route B (replacement):** introduce flavor-conversion terms derived from the P field and describe directly the β -decay Q value, transition rate, and half-life.

Note (moved to the discussion section): the theoretical mechanism of Route B, the W/Z mass reference points from the two-component coupled Q-ball, and the final numerical judgment of acceptance/rejection are handled separately in Section 3.2.

To avoid ending the theory-side reference points at the level of names only, this section also fixes the concrete route leading to the W/Z mass reference points. On the theory side of this paper, starting from the non-transverse basis $\{\delta P_0, \delta P_L\}$ after extracting light, we assume a two-component coupled Q-ball and scan the branches of the same family by a four-state shooting method for the state vector $y = (f_0, f'_0, f_L, f'_L)$. Along this route, the initial full scan reconstructs 41 localized sectors and a strict vector table of 905,838 lines, and in the final organization the W/Z mass reference points belonging to the same family (17, 1, 1) are fixed under the coupled-localization condition. Specifically, $\kappa_{\text{coupled},W} = 1.6024037199$ and $\kappa_{\text{coupled},Z} = 1.6817234304$ are both positive, and the relative errors of the reference points are 2.40×10^{-5} and 2.07×10^{-5} (about 0.002 percent). Therefore the theoretical footing of the weak-interaction sector is not blank, and the older single-component zero- κ truncation is treated not as a physical rejection rule but as an artifact of component decomposition.

The corresponding public figure, `output/public/quantum/v2_trial3_weak_checkpoint_summary.pdf`, summarizes the coupled-localization condition for the W/Z mass reference points, the reference-point errors for family (17, 1, 1), positive coupled localization, the numerical solution method and spectral scale, and the extension of the charge window.

Supplement: this figure is a summary of the theory-side reference points. The observational acceptance/rejection of the weak interaction itself is left to Section 3.2 and the numerical audits of Part III-B as the canonical source. [1]

The minimal unified expression is

$$i\hbar \frac{d}{dt} \Psi_f = \left(H_{\text{EW}} + \epsilon_f H_{\text{flavor}}^{(P)}[P, \nabla P, \partial_t P] \right) \Psi_f \quad (10)$$

where $\epsilon_f = 0$ corresponds to Route A and $\epsilon_f \neq 0$ to Route B.

The observational interface for β decay is fixed as

$$Q_\beta^{(P)} = Q_\beta^{(\text{EW})} + \Delta Q_\beta^{(P)}, \quad \Gamma_\beta^{(P)} = \Gamma_\beta^{(\text{EW})} (1 + \eta_\beta^{(P)}) \quad (11)$$

and ΔQ_β and η_β are treated as audit quantities for the "P-added effect."

Required checks (minimum requirements fixed in this paper):

1. **Definition of the flavor-conversion mechanism:** make explicit the degrees of freedom, symmetry, and conservation laws of $H_{flavor}^{(P)}$.
2. **Decision between transplant/replacement strategies:** fix on an observable basis which of Route A/B is the main system.
3. **Fixing the observational gate:** audit Q_β , half-life, $\log ft$, and if necessary CKM/PMNS consistency under the same interface.
4. **Explicit rejection conditions:** do not adopt any route that fails to satisfy the 3σ operating threshold.

Until these four conditions are closed, the observational acceptance/rejection of the weak-interaction sector remains on watch.

3 Discussion

This chapter crosses the theoretical sections 2.5-2.8 and organizes what has been fixed and what still remains as a follow-up task.

3.1 Quantum Foundations Fixed in This Paper

The point reached in Part III-A is not a declaration that quantum mechanics as a whole has been replaced. Rather, it is that the foundational entry points for which the P-model takes responsibility have been fixed explicitly. Specifically, this paper has fixed: (i) the minimum conditions under which the phase of the time wave reduces to the Schrödinger / KG forms, (ii) the derivation of the Born rule through phase diffusion and linear detection response, (iii) the coarse-grained Lueders/Kraus update through decoherence and conditional update, (iv) the transverse photon branch and the selection rule for the fine-structure constant, and (v) the nonseparable pair amplitude and the source dynamics of three-wave mixing.

Electromagnetism, however, must be read on two layers. On the light side, the transverse photon branch and the massless light mode are fixed, along with the selection rule that chooses a single fine-structure constant from a single frozen action and its reference value. By contrast, the origin of local U(1) redundancy, charge quantization, and charge normalization is still not closed from P alone, so on the observational side the already adopted U(1) sector is retained.

3.2 Residual Items, Conditional Holds, and the Division of Labor with Part III-B

The items remaining after the reorganization are not all unresolved in the same sense. First, for the fine-structure constant, the central theory-side claim is already firm. The roughly 1.93 percent gap seen in the old spherical approximation is a diagnostic value, and in the current representative value the gap has already been reduced to about 0.00624 percent below $1/137$. The remaining four-dimensional shape factors, operator corrections, and the bridge to observables are treated not as failure of the theory, but as follow-up tasks on the observable side. Accordingly, Part III-A retains the selection rule and the reference value as the canonical source, while local tables and auxiliary diagnostics are left to the supporting surfaces of Part III-B / Part IV.

Second, the origin of local $U(1)$ redundancy, charge quantization, and charge normalization remains a truly unresolved item. The transverse photon branch, the massless light mode, and the selection rule for the fine-structure constant are already fixed, but the full local source theorem including the adopted $U(1)$ sector is still not closed. Here Part III-A makes the missing theory-side piece explicit, while Part III-B takes as canonical the verification surface using the already adopted sector.

Third, the weak interaction must be read with a separation between the theoretical mechanism and the observational acceptance/rejection. On the theory side, the W/Z mass reference points on the same family generated by the two-component coupled Q-ball have been fixed on family $(17, 1, 1)$ under the coupled-localization condition, with $\kappa_{\text{coupled},W} = 1.6024037199$, $\kappa_{\text{coupled},Z} = 1.6817234304$, and relative reference-point errors $= (2.40, 2.07) \times 10^{-5}$. Therefore the weak-interaction sector does not remain without any footing at all. On the other hand, the paper's acceptance/rejection judgment takes the numerical audit of Part III-B as canonical, and at the present stage is fixed as adopting Route A and rejecting Route B. Thus the V-A mechanism of Route B is retained as a theoretical candidate, but not as an adopted operating mechanism.

Accordingly, Part III-A is the document that gives the fixed points, residual gaps, and conditionally held items on the theory side, while the verification results based on public primary data, the cross-sectional scoreboard, differential predictions, and the falsification-condition pack are canonical in Part III-B. The reproduction commands, fixed artifact list, and audit gates are canonical in Part IV (verification materials). [1]

4 Conclusion

The conclusions of this paper can be summarized in five points.

1. The minimum conditions connecting the local phase of P and the proper-time mapping to the Schrödinger / KG forms have been fixed.
2. For the Born rule, the detection probability $p \propto |\psi|^2$ has been effectively derived through phase diffusion and linear detection response under the four conditions of phase mixing, linear detection response, weak backreaction, and frequentist coarse-graining.
3. For measurement, diagonal Lueders/Kraus update has been recovered in coarse-grained form from the detector's stable basis, decoherence, and conditional update.
4. For electromagnetism, the transverse photon branch and the selection rule for the fine-structure constant have been fixed, while the origin of local U(1) and charge normalization remain unresolved.
5. For entanglement, the nonseparable pair amplitude, the source dynamics of three-wave mixing, and the audit surface toward Bell datasets have been fixed, while the judgment itself is left to Part III-B.

For the weak interaction, the W/Z mass reference points from the two-component coupled Q-ball are retained as theory-side references, while the canonical source for observational acceptance/rejection is the numerical audit of Part III-B, and the present conclusion is fixed as adopting Route A and rejecting Route B.

Accordingly, the role of Part III-A is to fix the theoretical foundations of the quantum volume, while the final judgment table that decides theoretical acceptance or rejection is left to the verification system of Part III-B / Part IV.

5 Reproducibility / Availability

The theoretical formulas, fixed artifacts, and reproduction commands referenced in this paper are collected in Part IV (verification materials). For public artifacts, fixed parameters, audit summaries, and reproducibility routes, refer to the Part IV ledger and the public artifacts. [1]

The verification results based on public primary data, the cross-sectional judgment tables, the differential predictions, and the falsification-condition pack are canonical in Part III-B. Part III-A is positioned as the document that provides the theory-side fixed points and the interpretive boundaries used in that verification system.

6 Appendix A: Data Sources (Primary Sources)

This appendix is a concise list of representative primary sources referenced in the theoretical sections of Part III-A. Detailed URLs, access dates, and the artifact correspondence table are canonical in the Part IV ledger. [1]

Topic	Representative primary sources	Main text sections
Bell tests	NIST Bell test data, Delft event-ready public primary data	2.6.4, 3.2
α / atomic metrology	atom-recoil α , electron $g - 2$, H 1S-2S	2.6.1-2.6.3
Gravity x quantum interference	COW, atom interferometer	2.5, 3.1
Optical / single-photon interference	single-photon interference, optical reference systems	2.6.4

Bell test sources: [3][4][5]

α / atomic metrology sources: [8][9][11][12]

Gravity x quantum interference sources: [6][7]

Optical / single-photon interference sources: [10][2]

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